Signed Friedkin–Johnsen Models: Opinion Dynamics With Stubbornness and Antagonism

Muhammad Ahsan Razaq and Claudio Altafini, *Senior Member, IEEE*

***Abstract*—The aim of this article is to investigate how opinions propagate in presence of stubborn behavior (i.e., agents that tend to defend their stance) and of signed (i.e., antagonistic) interactions among the agents. The stability properties of the resulting system, denoted signed Friedkin–Johnsen (SFJ) model, are studied in detail, and also the convergence to consensus for a concatenation of such SFJ models. A noteworthy feature of the SFJ model is that the opinion dynamics need not be confined to the convex hull of the initial conditions, meaning that cleavage can increase because of the antagonistic interactions. In particular, we show that the concatenated SFJ model can behave as a nonnormal system, in which the opinions transiently distance themselves only to rejoin a common consensus value asymptotically.**

***Index Terms*—Consensus, opinion dynamics, signed graphs.**

I. Introduction

OPINION dynamics models aim to study how the opinions and beliefs of individuals change over time with the influence of other individuals' opinion within a group or a community [1], [2]. These models have been studied extensively across different fields, including sociology [3], [4], psychology [5], finance [6], and political science [7] as they help us understand how social norms and behaviors are formed and evolve over time. The models work by modifying the opinions (i.e., the state variables) of the agents based on the previous values of the agents themselves and on the opinions of their neighbors.

Most of the opinion dynamics models currently used in the literature rely on some form of averaging to update the opinion of an agent. In the classical DeGroot model, for instance, an agent updates its opinion by taking the weighted average of the opinions of its neighbors on a graph of interaction [8]. In a Friedkin–Johnsen (FJ) model, a convex combination of the averaging part and of the initial condition of the agent is instead considered, as a way to express the stubbornness of an agent, i.e., its reluctance to change opinion [9]. In a bounded confidence model, the averaging rule is instead applied only to the agents that have an opinion similar to that of the agent under consideration [10].

As a consequence of the averaging rule, the opinions of the agents tend to get closer to each other, i.e., the opinions are always confined inside the convex hull of the initial conditions, where they evolve according to the specific dynamical rules imposed by the model, e.g., toward consensus for the DeGroot model, or toward clusters of identical values for the bounded confidence model. The endpoint of an FJ model instead typically consists of all different opinions ("cleavage" phenomenon [4]), which are nevertheless closer to each other than the initial conditions. As a consequence, when multiple FJ models are concatenated in a sequential way as we did in [11], then the contraction in opinion space at each step leads eventually to consensus. The intuition behind the concatenated FJ model of [11] is that it can represent a multistage discussion, in which the endpoint of the kth discussion becomes the initial condition of the $(k+1)$st discussion. See [12] for an application of this framework to climate change negotiations.

The convex hull invariance property common to all models described so far is reasonable in a context of collaboration among the agents. However, it cannot capture real-life phenomena like opinion polarization, which in fact is typically modeled using some form of nonlinear dynamics [13], [14]. If we want to escape the convex hull of the initial conditions using linear models, the only possibility is to include some form of antagonism in the interactions, i.e., to use signed graphs. Signed graphs, in fact, can effectively represent a form of "repulsive influence" among interacting individuals, and are a suitable way to capture polarization. It is already clear from [15] that in some cases, namely when the signed graph is structurally balanced, a signed DeGroot model converges to bipartite consensus, which is already potentially violating the convex hull invariance. A similar argument was recently developed for FJ models on signed graphs in [16] and [17]. However, for the form of Laplacian considered in these papers (the so-called "opposing Laplacian" in the terminology of [18]) the structurally balanced condition is strictly necessary: as soon as the signed graph becomes unbalanced the (negated) Laplacian becomes Hurwitz, and all opinions collapse to 0, which is not a practically interesting situation for an opinion dynamics model.

The aim of this article is to propose an alternative formulation for an FJ opinion dynamics model on a signed graph, bypassing the bottleneck of needing a structurally balanced signed graph. Our formulation relies on a second form of signed Laplacian

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The authors are with the Division of Automatic Control, Department of Electrical Engineering, Linköping University, SE-58183 Linköping, Sweden (e-mail: muhammad.ahsan.razaq@liu.se; claudio.altafini@liu.se).

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(called "repelling Laplacian" in [18]), which does not require structural balance in order to violate the convex hull invariance property. The main properties of a "repelling" Laplacian are that it has always zero as an eigenvalue but may not be stable [19], or may be stable but may fail to converge [20]. These make our model much more challenging to analyze than the one based on the opposing Laplacian, and its behavior much richer (the structurally balanced case becomes a special case). Similar "repelling" adjacency matrices can be considered as replacement for stochastic matrices when dealing with discrete-time (DT) opinion dynamics models. The FJ model on signed graphs we introduce in this article is henceforth denoted *signed FJ* (SFJ), and its concatenated counterpart *concatenated SFJ*. Achieving stability or instability in this model is intrinsically a collective behavior, which single agents, having only local information on their neighbors, cannot anticipate (or control). In this respect, the SFJ model exhibits a more realistic behavior than averaging models, where the global opinion dynamics is qualitatively always the same (i.e., convex hull invariance and convergence always hold).

To analyze the behavior of our SFJ models, we use the tools developed in [20], in particular the notions of eventually stochastic (ES) matrices (for DT SFJ) and eventually exponentially positive (EEP) Laplacians (for continuous-time SFJ). By itself, choosing an ES matrix (or an EEP Laplacian) is however not enough to guarantee stability of the SFJ, as easily shown in counterexamples. A first sufficient condition for stability, investigated in [21], is that the interaction matrix is normal (plus ES or EEP). In this article, inspired by [22], we sharpen this sufficient condition to the condition that the numerical range of an ES signed interaction matrix has a sharp point at 1 and all other boundary points inside the unit circle (respectively, the numerical abscissa of the signed Laplacian L has a sharp point at 0, with $-L$ marginally stable). Such condition (which subsumes normality as a special case) guarantees that the SFJ model is globally asymptotically stable, and hence a unique equilibrium point always exists for it. Furthermore, the same condition is sufficient to guarantee that the concatenated SFJ model converges to consensus over an infinite number of discussion events.

At the same time, the convex hull invariance property can be violated, as easily shown by examples. In particular this happens when the ES matrix that describes the entire evolution of the SFJ model (which can be easily obtained in closed form) has negative entries. This effect is more pronounced when it is not normal, and in particular, when it has a numerical radius larger than one (its spectral radius is one, as it is an ES matrix). The opinions can therefore evolve to become more distant from each other than they were at the beginning of a discussion. We believe that this is a novel feature for linear models of opinion dynamics (again excluding the cases of opposing Laplacians which require structural balance of the interaction graph).

A consequence is that in a concatenated SFJ model, we can observe unusual phenomena, such as 1) on each discussion we have a unique equilibrium point, but the opinions become progressively more distant as we concatenate multiple discussions; 2) asymptotically (in the number of discussion being concatenated) consensus is achieved (as in the unsigned concatenated FJ model of [11]), but transiently we register an increasing opinion cleavage, i.e., opinions that temporarily distance themselves, before rejoining a common consensus value. Both phenomena are not documented in the literature on opinion dynamics, and thereby enrich the number of possible behaviors exhibited by opinion dynamics models. The latter situation is analogous to what has been emphasized in a lot of recent literature on nonnormal dynamics over networks [23], [24], [25], and is historically well-known also in fields like ecology under the name of reactivity [26].

The rest of this article is organized as follows. Section II gives preliminary concepts on signed graphs and matrix theory. Section III introduces our SFJ models in continuous and DT, and reviews the properties of ES and EEP matrices. Section IV provides sufficient conditions for asymptotic stability of the SFJ models, and Section V extends the results to concatenated SFJ models, bipartite consensus, and time varying concatenated SFJ models. Section VI gives a series of examples on a small-scale social network. Finally, Section VII concludes this article. All proofs are in the Appendices.

With respect to the conference paper [21], the conditions we present in this manuscript are significantly more general: in [21] all stubbornness coefficients have to be positive while here they are not; Ref. [21] dealt only with the case of normal interaction matrices, while the conditions based on sharp points for the numerical range considered here are much broader. Also the extensions to bipartite and time-varying concatenated SFJ are novel material.

II. Preliminary Material

Notations: Let $\mathbb{R}$, $\mathbb{R}_{\geq 0}$, and $\mathbb{C}$ be the sets of real numbers, nonnegative real numbers, and complex numbers. Real numbers are denoted by lowercase letters $a, b, c, \ldots$ while bold lowercase letters $\mathbf{x}, \mathbf{y}, \mathbf{z}, \ldots$ represent vectors of length n in $\mathbb{R}^n$. $\mathbf{0}$ and $\mathbb{1}$ are vectors of entries 0 and 1, respectively. Matrices are represented by capital Latin or Greek letters $X, Y, \Theta, \ldots$. Given a matrix $Q \in \mathbb{R}^{n\times n}$ of entries Q_{ij}, Q^T denotes its transpose. For $Q \in \mathbb{C}^{n\times n}$, Q^* represents its complex conjugate transpose. The eigenvalues of the matrix Q are denoted by $\lambda(Q)$, and the spectrum as $\Lambda(Q) = \{\lambda_1(Q), \ldots, \lambda_n(Q)\}$. The spectral radius and spectral abscissa of Q are defined as $\rho(Q) = \max_{i=1,\ldots,n} |\lambda_i(Q)|$ and $\mu(Q) = \max_{i=1,\ldots,n} \Re(\lambda_i(Q))$, where $\Re(\cdot)$ and $\Im(\cdot)$ denote the real and imaginary part of a complex number. The matrix I represents an identity matrix of appropriate dimensions. $\mathbb{S}^{n-1} \subset \mathbb{R}^n$ denotes a unit sphere in $\mathbb{R}^n$. For a vector $\mathbf{x} \in \mathbb{R}^n$, the diagonal matrix $\mathrm{diag}(\mathbf{x}) \in \mathbb{R}^{n\times n}$ has the entries of $\mathbf{x}$ on the diagonal. The p-norm of a vector $\mathbf{x} \in \mathbb{R}^n$ is denoted $\|\mathbf{x}\|_p$ and is defined as $\|\mathbf{x}\|_p = (|\mathbf{x}_1|^p + |\mathbf{x}_2|^p + \cdots + |\mathbf{x}_n|^p)^{\frac{1}{p}}$.

A. Matrix Theory

If the eigenvalues of a square matrix Q have $\Re(\lambda(Q)) < 0$ (respectively, $|\lambda(Q)| < 1$), then the matrix is called Hurwitz (respectively, Schur) stable. It is marginally stable if $\Re(\lambda(Q)) \leq 0$ (respectively, $|\lambda(Q)| \leq 1$), and the eigenvalues such that $\Re(\lambda(Q)) = 0$ (respectively, $|\lambda(Q)| = 1$) are simple eigenvalues,

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i.e., have multiplicity 1. If there exists a permutation matrix P such that $P^TQP = \begin{bmatrix} Q_1 & Q_2 \\ 0 & Q_3 \end{bmatrix}$ where Q_1 and Q_3 are nontrivial square matrices, then the matrix Q is said to be reducible. It is irreducible if it is not reducible.

A matrix Q that has all its entries $Q_{kl} > 0$ is said positive and denoted $Q > 0$; it is said nonnegative if $Q_{kl} \geq 0$ and it is denoted $Q \geq 0$. The corank of a matrix Q is defined as the dimension of the kernel space of Q, $\ker(\mathrm{Q})$. The matrix Q is normal if $QQ^T = Q^TQ$. Positive definiteness (respectively, positive semi definiteness) of a matrix Q is defined as $\mathbf{x}^TH(Q)\mathbf{x} > 0$ (respectively, $\mathbf{x}^TH(Q)\mathbf{x} \geq 0$) for all $\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0$ (respectively, $\mathbf{x} \in \mathbb{R}^n$) where $H(Q) = (Q + Q^*)/2$, and it is denoted $Q \succ 0$ (respectively, $Q \succeq 0$). Negative definiteness (respectively, semi-definiteness) is denoted $Q \prec 0$ (respectively, $Q \preceq 0$) and obviously corresponds to $-Q \succ 0$ (respectively, $-Q \succeq 0$). The skew-Hermitian part of the matrix Q is denoted $S(Q) = (Q - Q^*)/2$. The spectral norm of a matrix Q is defined as $\|Q\|_2 = \max_{\|\mathbf{x}\|=1} \|Q\mathbf{x}\|_2$ and is attained when $\mathbf{x}$ is the eigenvector corresponding to the maximum eigenvalue of Q^TQ. The norm can also be defined as $\|Q\|_2 = \max_{\mathbf{x},\mathbf{y}\in\mathbb{S}^{n-1}} |\mathbf{x}^TQ\mathbf{y}|$ where the vectors $\mathbf{x}$ and $\mathbf{y}$ can be easily found using singular value decomposition.

B. Signed Graphs

A directed graph (digraph) with n vertices $\mathcal{V} = \{v_1, v_2, \ldots, v_n\}$ and an edge set $\mathcal{E}$ is represented by $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$ where an edge $(k, l) \in \mathcal{E}$ is a link from vertex v_k to v_l and the weighted adjacency matrix A represents the interaction pattern in the graph, with $A_{kl} = 0$ *iff* $(l, k) \notin \mathcal{E}$. For signed graphs, the values of A_{kl} can be positive or negative depending on the relationship between the vertices ("friends" or "enemies"). For an undirected graph, we have $A^T = A$, while a digraph is called weight balanced if $A\mathbb{1} = A^T\mathbb{1}$.

For a signed graph, the weighted in-degree vector is defined as $\xi_{\text{in}} = [\sum_{j=1,i\neq j}^{n} A_{ij}]_{n\times 1} \in \mathbb{R}^n$. The "repelling Laplacian" [18] is defined as $L = \Sigma_{\text{in}} - A$ where $\Sigma_{\text{in}} = \text{diag}(\xi_{\text{in}})$. Notice that by construction $L\mathbb{1} = \mathbf{0}$.

A path in the graph is defined as collection of edges $(v_k, v_1), (v_1, v_2), \ldots, (v_{r-1}, v_r), (v_r, v_l)$ such that $A_{lr}A_{r(r-1)} \ldots A_{21}A_{1k} \neq 0$. If vertex i can reach any other vertex in the graph via a directed path, then a directed spanning tree rooted at vertex i exists. If every pair of vertices in the graph is connected through a directed path, then the graph is said strongly connected.

C. Eventual Positivity

In linear algebra, a (signed) matrix $Q \in \mathbb{R}^{n\times n}$ is said to have the Perron–Frobenius property (denoted $Q \in \mathcal{PF}$) if $\rho(Q)$ is a simple real positive eigenvalue such that $\rho(Q) > |\lambda(Q)|$ for all $\lambda \in \Lambda(Q), \lambda \neq \rho(Q)$, and the corresponding right eigenvector is positive, see [27]. A general class of matrices having the Perron–Frobenius property is given by the class of so-called *eventually positive (EP)* matrices. The following definitions and properties can be found in [20].

Definition 1 [Eventually Positive (EP)]: The matrix $Q \in \mathbb{R}^{n\times n}$ is said EP if $\exists$ an integer k_o such that $Q^k > 0$ for $k \geq k_o$.

Following, e.g., [27], to denote an EP matrix Q we use the notation $Q \overset{\vee}{>} 0$.

Lemma 1 (See [27]): For a matrix $Q \in \mathbb{R}^{n\times n}$, the following properties are equivalent 1) Q, $Q^T \in \mathcal{PF}$; 2) $Q \overset{\vee}{>} 0$; 3) $Q^T \overset{\vee}{>} 0$.

Definition 2 [Eventually Exponentially Positive (EEP)]: The matrix $Q \in \mathbb{R}^{n\times n}$ is said EEP if $\exists\, d \in \mathbb{R}_{\geq 0}$ such that $Q + dI \overset{\vee}{>} 0$.

An equivalent characterization of EEP is that $e^{Qt} > 0\ \forall\, t > t_o$, for some $t_o \in \mathbb{R}_{\geq 0}$.

D. ES Matrices

Definition 3 [Eventually Stochastic (ES)]: The matrix $Q \in \mathbb{R}^{n\times n}$ is said ES if $Q \overset{\vee}{>} 0$ and $Q\mathbb{1} = \mathbb{1}$. If in addition it is $\mathbb{1}^TQ = \mathbb{1}^T$, then Q is said eventually doubly stochastic (EDS).

Lemma 2 (See [28]): For an ES matrix Q, $1 \in \Lambda(Q)$ is a simple and strictly dominant eigenvalue, and the right and left eigenvectors $\mathbf{v}$ and $\mathbf{w}$ corresponding to the eigenvalue 1 are positive.

E. Signed Eventually Positive and Stochastic Matrices

A (signed) matrix $Q \in \mathbb{R}^{n\times n}$ is said to have the signed Perron–Frobenius property (denoted $Q \in \mathcal{SPF}$) if $\rho(Q)$ is a simple real positive eigenvalue such that $\rho(Q) > |\lambda(Q)|$ for all $\lambda \in \Lambda(Q)$, $\lambda \neq \rho(Q)$, and the corresponding right eigenvector has all nonzero components.

Lemma 3 (See [28]): For a matrix $Q \in \mathbb{R}^{n\times n}$, the following are equivalent:

1) Q, $Q^T \in \mathcal{SPF}$, and the right and left eigenvectors $\mathbf{v}$ and $\mathbf{w}$ of Q relative to $\rho(Q)$ are such that $\mathbf{v}_i\mathbf{w}_i < 0$, or $\mathbf{v}_i\mathbf{w}_i > 0\ \forall i = 1, \ldots, n$.
2) $\exists\, \Omega = \text{diag}(\boldsymbol{\sigma})$ such that $\Omega Q \Omega \overset{\vee}{>} 0$, where $\boldsymbol{\sigma} = [\sigma_1 \quad \ldots \quad \sigma_n]^T$, and $\sigma_i = \pm 1$.

The diagonal signature matrix $\Omega = \text{diag}(\boldsymbol{\sigma})$ is sometimes called a gauge transformation.

Definition 4: A matrix $Q \in \mathbb{R}^{n\times n}$ is said

1) signed eventually positive (SEP) if $\exists\, \Omega = \text{diag}(\boldsymbol{\sigma})$ such that $\Omega Q \Omega \overset{\vee}{>} 0$,
2) signed eventually exponentially positive (SEEP) if $\exists\, \Omega = \text{diag}(\boldsymbol{\sigma})$ and $\exists\, d \in \mathbb{R}_{\geq 0}$ such that $\Omega Q \Omega + dI \overset{\vee}{>} 0$, and
3) signed eventually stochastic (SES) if $\exists\, \Omega = \text{diag}(\boldsymbol{\sigma})$ such that $\Omega Q \Omega \overset{\vee}{>} 0$, $\Omega Q \Omega \mathbb{1} = \mathbb{1}$.

Lemma 4: For an ES matrix Q, $Q - I$ is EEP and $\mu(Q - I) = 0$ is simple.

The proof is in Appendix A.

F. Numerical Range

The following definitions and properties can be found in [29].

Definition 5 (Numerical range): The numerical range of a matrix $Q \in \mathbb{C}^{n\times n}$ is defined as a contour in $\mathbb{C}$ given by $\omega(Q) = \{\mathbf{x}^*Q\mathbf{x} : \mathbf{x} \in \mathbb{C}^n, \|\mathbf{x}\| = 1\}$.

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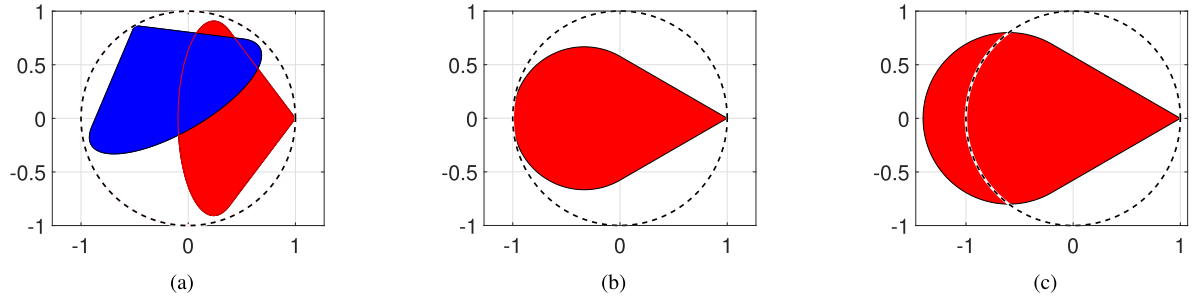

Fig. 1. Sharp point in $\omega(Q)$ and its properties. In each panel the dashed circle is $\rho(Q) = 1$ and the solid area $\omega(Q)$. (a) Sharp point is strictly dominant, $\omega(Q)$ (red) and $\omega(e^{i2\pi/3}Q)$ (blue). (b) Sharp point is not strictly dominant, $H(Q)$ has eigenvalues at ± 1. (c) Sharp point at 1, but $r(Q) > 1$.

The numerical range of a matrix $Q \in \mathbb{C}^{n\times n}$, denoted $\omega(Q)$, constitutes a compact and convex set in $\mathbb{C}$ encompassing all eigenvalues of Q, i.e., $\Lambda(Q) \subset \omega(Q)$. The boundary points of the convex set $\omega(Q)$ are denoted $\partial\omega(Q)$. The projection of $\omega(Q)$ onto the real axis is represented as $\Re(\omega(Q))$ and is equivalent to $\omega(H(Q))$. Similarly, its projection onto the imaginary axis, $\Im(\omega(Q))$, equals $\omega(S(Q))$. For matrices $Q_1, Q_2 \in \mathbb{C}^{n\times n}$, the eigenvalues of their sum obey to the relationship: $\Lambda(Q_1 + Q_2) \subset \omega(Q_1 + Q_2) \subset \omega(Q_1) + \omega(Q_2)$. Furthermore, If $Q_2 \succeq 0$, then $\Lambda(Q_1Q_2) \subset \omega(Q_1)\omega(Q_2)$ and if $0 \notin \omega(Q_2)$, then $\Lambda(Q_1Q_2^{-1}) \subset \omega(Q_1)/\omega(Q_2)$ where the sum, product, and division of sets is done element-wise, see [29].

Definition 6 (Numerical radius): The numerical radius of a matrix $Q \in \mathbb{C}^{n\times n}$ is defined as $r(Q) = \max\{|z| : z \in \omega(Q)\}$.

Definition 7 (Numerical abscissa): The numerical abscissa of a matrix $Q \in \mathbb{C}^{n\times n}$ is defined as $\mu(H(Q)) = \max\{\Re(z) : z \in \omega(Q)\}$.

Recall that in general for a matrix Q, if $\rho(Q)$ is an eigenvalue, the following relationship holds:

$$\rho(Q) \le \rho(H(Q)) \le r(Q) \le \|Q\|_2.$$

For any matrix Q, the greater the disparity between $\rho(Q)$ and $r(Q)$ (or $\mu(Q)$ and $\mu(H(Q))$), the more transient behavior can be observed in the system dynamics, whether discrete $\mathbf{x}(k+1) = Q\mathbf{x}(k)$ or continuous $\dot{\mathbf{x}}(t) = Q\mathbf{x}(t)$. Moreover, larger differences indicate that the matrices Q deviate more from normality [25].

Definition 8 (Radial matrix [30]): A matrix $Q \in \mathbb{C}^{n\times n}$ is called radial if its spectral radius is equal to its spectral norm, i.e., $\rho(Q) = \|Q\|_2 \, (= r(Q))$.

The matrix Q is radial *iff* $\rho(Q)I_n - Q^TQ \succeq 0$. By construction, symmetric (Hermitian), skew-symmetric (skew-Hermitian), unitary, and normal matrices are examples of radial matrices. A "sharp point" of a convex set is a boundary point that acts as a "corner," implying nonunique tangents.

Definition 9 (Sharp point [29]): For a matrix $Q \in \mathbb{C}^{n\times n}$, a point $\zeta \in \partial\omega(Q)$ is called a sharp point of $\omega(Q)$ if there are angles ϑ_1 and ϑ_2 with $0 \le \vartheta_1 < \vartheta_2 < 2\pi$ such that, for all $\vartheta \in (\vartheta_1, \vartheta_2)$, the following holds:

$$\Re(e^{i\vartheta}\zeta) = \max\{\Re(\beta) : \beta \in \omega(e^{i\vartheta}Q)\}.$$

In words, at a sharp point ζ, the real part of $e^{i\vartheta}\zeta$ is equal to the maximum value over the real part of the numerical range $\omega(e^{i\vartheta}Q)$ for all ϑ in the specified range $(\vartheta_1, \vartheta_2)$. A sharp point $\zeta \in \partial\omega(Q)$ is an eigenvalue of Q. For a matrix $Q \in \mathbb{R}^{n\times n}$, if $\rho(H(Q)) = \rho(Q)$ is simple and strictly dominant (respectively, $\mu(Q) = \mu(H(Q))$ is simple), it implies the existence of a sharp point in $\omega(Q)$ at $\rho(Q)$ (respectively, $\mu(Q)$).

Lemma 5 (See [29]): For a matrix $Q \in \mathbb{C}^{n\times n}$, the following statements are equivalent: $r(Q) \le 1$; $\rho(H(e^{i\vartheta}Q)) \le 1$; $\mu(H(e^{i\vartheta}Q)) \le 1, \forall\vartheta \in \mathbb{R}$.

The numerical range of the product $e^{i\vartheta}Q$, i.e., $\omega(e^{i\vartheta}Q)$, is the same as $\omega(Q)$ rotated around the origin by an angle ϑ. In Fig. 1(a), the numerical range of an ES matrix Q, i.e., $\omega(Q)$ with a sharp point at 1, and $\omega(e^{i\vartheta}Q)$ where $\vartheta = 2\pi/3$ are shown.

Lemma 6 ([22, Lemma 3.10]): For a matrix $Q \in \mathbb{R}^{n\times n}$ and $\lambda \in \mathbb{R}, \lambda \in \omega(Q)$ is a sharp point *iff* the following two conditions are satisfied:

1) $\lambda \in \Lambda(Q)$ and $\lambda \in \partial\omega(Q)$.
2) Q and $H(Q)$ have a common eigenspace corresponding to the eigenvalue λ.

The above lemma implies that the left and right eigenvector associated to the eigenvalue $\lambda \in \mathbb{R}$ of a matrix Q are the same if $\lambda \in \omega(Q)$ is a sharp point. Also for sharp points we can have a notion of strict dominance.

Definition 10 (Strictly dominant sharp point): For a matrix $Q \in \mathbb{C}^{n\times n}$, a point $\zeta \in \partial\omega(Q)$ is called a strictly dominant sharp point of $\omega(Q)$ if $\zeta \in \omega(Q)$ is a sharp point and $|\zeta| > \varphi \; \forall \varphi \in \partial\omega(W), \varphi \ne \zeta$.

The meaning of the previous definition is that the sharp point ζ is strictly larger in modulus than any boundary point $\partial\omega(Q)$ at any rotation angle. Fig. 1(a) shows $\omega(Q)$ with a strictly dominant sharp point at 1 (red) and $-0.05 + 0.866i$ (blue). Strict dominance of a sharp point can be considered an extension to signed matrices of the idea of aperiodicity for nonnegative matrices. The following lemma (proof in Appendix A) shows that the matrix Q retains its numerical range after a gauge transformation.

Lemma 7: For a matrix $Q \in \mathbb{R}^{n\times n}$ and a gauge transformation matrix Ω, $\Omega = \mathrm{diag}(\boldsymbol{\sigma})$, $\boldsymbol{\sigma} = [\sigma_1 \;\; \ldots \;\; \sigma_n]^T$, and $\sigma_i = \pm 1$, it is $\omega(\Omega Q \Omega) = \omega(Q)$.

III. Problem Formulation

A. SFJ Model

1) Discrete-Time SFJ: According to [31], in DT, the FJ model is defined as the convex combination

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$$\mathbf{x}(t+1) = (I-\Theta)\,W\mathbf{x}(t) + \Theta\mathbf{x}(0), \tag{1}$$

where $\mathbf{x}(t) \in \mathbb{R}^n$ defines the state ("opinion") of n agents, $\Theta = \mathrm{diag}([\theta_1, \ldots, \theta_n])$ is the diagonal matrix containing the stubbornness coefficients $\theta_i \in [0,\, 1)$ of the agents (i.e., their "attachment" to their initial opinions $\mathbf{x}(0)$), and W is the weighted adjacency matrix representing the communication graph $\mathcal{G}$ of the agents. In this work, we assume that the agents can have antagonistic interactions, and hence that W is a signed weighted adjacency matrix. In the case that no agent is stubborn, i.e., when $\Theta = 0_{n\times n}$, the model (1) becomes

$$\mathbf{x}(t+1) = W\mathbf{x}(t), \tag{2}$$

and we want this special case to behave as a DeGroot model, namely to achieve consensus. In order to obtain this for signed graphs, we need to impose conditions on W. In particular, we will typically work with W which is a ES matrix, as per Definition 3. In fact, for ES matrices, a series of properties were observed in [20] and summarized in [21, Lemma 5]. With respect to this lemma, the formulation given below in Lemma 9 is enhanced to include W with a sharp point at 1 in $\omega(W)$, which significantly increases the domain of possible matrices W. The proof is omitted, as it is similar to the proof in [22, Thm. 4.2]: for W ES with a sharp point in $\omega(W)$ at 1, we get $W - I$ EEP with a sharp point in $\omega(W - I)$ at 0. Let us first give a characterization of ES matrices with a sharp point at 1 stating a Lemma with proof in Appendix A.

Lemma 8 (Sharp points and ES): If the matrix W is ES and $\omega(W)$ has a sharp point at 1, then W is EDS and hence it is weight balanced.

Remark 1: According to Lemma 8, an ES matrix W with $1 \in \omega(W)$ having a sharp point is EDS; but the sharp point may not be strictly dominant as can be seen in Example 1.

Example 1: The matrix

$$Q = \begin{bmatrix} -0.2 & 0.4 & -0.4 & 1.2 \\ 0.4 & -0.2 & 1.2 & -0.4 \\ 0.4 & 0.4 & -0.2 & 0.4 \\ 0.4 & 0.4 & 0.4 & -0.2 \end{bmatrix}$$

is EDS and $\omega(Q)$ has a sharp point at 1, but $r(Q) = 1.4$. The numerical range is plotted in Fig. 1(c). ■

Notice that in unsigned strongly connected and weight balanced digraphs the interaction pattern W always has 1 as a sharp point in $\omega(W)$.

Lemma 9 (Enhanced from [21, Lemma 5]): Consider the DT system (2), where the signed matrix W is such that $W\mathbb{1} = \mathbb{1}$. Consider the following conditions.

i) The system (2) achieves consensus.
ii) W is marginally Schur stable with a simple and strictly dominant eigenvalue $\rho(W) = 1$.
iii) W is ES.

Conditions (i) and (ii) are equivalent. Condition (iii) implies conditions (i) and (ii) but not viceversa. However, if in addition $\omega(W)$ has a sharp point at 1, then all three conditions are equivalent, W is EDS, and $H(I - W) \succeq 0$ with corank$(H(I - W)) = 1$. In addition, if W is radial, then $I - W^T W \succeq 0$ with a corank of 1.

2) Continuous-Time (CT) SFJ: A SFJ model can be built also in CT, for instance as follows:

$$\dot{\mathbf{x}}(t) = -\left((I-\Theta)\,L + \Theta\right)\mathbf{x}(t) + \Theta\mathbf{x}(0), \tag{3}$$

where $L = \Sigma_{\text{in}} - A$ is the signed Laplacian associated with the adjacency matrix A of the signed graph $\mathcal{G}$ and $\Theta = \mathrm{diag}([\theta_1, \ldots, \theta_n])$ represents again the stubbornness matrix. Also in this case, the requirement on L is that when $\Theta = 0$ the resulting system

$$\dot{\mathbf{x}}(t) = -L\mathbf{x}(t), \tag{4}$$

achieves consensus. For signed graphs, this is achieved when $-L$ is an EEP matrix. Similarly to the DT case, let us first discuss the property of an EEP matrix with a sharp point at 0.

Lemma 10 (Sharp points and EEP): If the matrix L is EEP and $\omega(L)$ has a sharp point at 0, then L is weight balanced.

The proof is in Appendix A. Notice that in unsigned strongly connected and weight balanced digraphs, the Laplacian matrix L always has 0 as a sharp point in $\omega(L)$.

Summarizing the results for signed Laplacian matrices from [20] and incorporating the existence of a sharp point at 0 in $\omega(L)$ [22, Thm. 4.2], we have the following results.

Lemma 11 ([20, Thm. 4, Lemma 4, Corollary 1],[22, Thm. 4.2]): Consider the CT system (4), where L is a signed Laplacian. Consider the following conditions.

i) The system (4) achieves consensus.
ii) $-L$ is marginally stable of corank 1.
iii) $-L$ is EEP.

Conditions (i) and (ii) are equivalent. Condition (iii) implies conditions (i) and (ii), but not vice versa. However, if in addition 0 is a sharp point of $\omega(L)$, then all three conditions are equivalent, $H(L) \succeq 0$ is of corank 1, and L is weight balanced.

IV. Convergence in SFJ Models

In this section, we study the asymptotic stability of models (1) and (3). In both cases, we need an assumption that some agents are partially stubborn, i.e., at least one agent has stubbornness value greater than 0 and always less than 1. In other words, we exclude the cases where any agent is totally stubborn ($\theta_i = 1$ for any i) and the case where no agent is stubborn ($\Theta = \mathbf{0}_{n\times n}$).

Assumption 1: At least one agent in the network is partially stubborn and no agent is fully stubborn, i.e., $\theta_i \in [0, 1)$ for any i, and $\Theta \neq \mathbf{0}_{n\times n}$.

A. Discrete Time Case

For the DT system (1), we need an assumption that W is ES with a strictly dominant sharp point on $1 \in \omega(W)$. In particular, we need to exclude the "periodic" case where $\omega(W)$ has a sharp point at 1 and touches -1 [see Fig. 1(b)]. In fact, in this case the symmetric part of W, i.e., $H(W)$, has eigenvalues ± 1, resulting in lack of convergence (the powers of $H(W)$ oscillate with periodicity 2).

Assumption 2: W is an ES matrix with $1 \in \omega(W)$ a strictly dominant sharp point.

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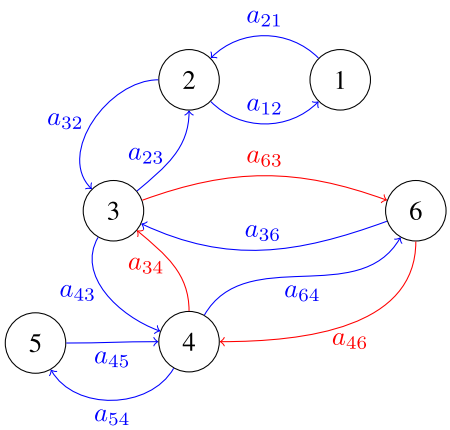

Fig. 2. Signed graph $\mathcal{G}$ representing a social network of six individuals. Blue edges are positive, and red edges are negative.

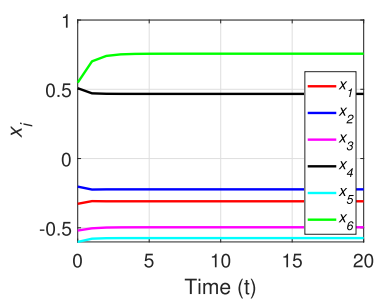

Fig. 3. Propagation of opinions for the single SFJ model of Fig. 2.

Remark 2: The condition that $1 \in \omega(W)$ is a strictly dominant sharp point is mathematically important, because it extends the idea of aperiodicity from nonnegative to signed matrices. It is also of practical relevance, as it leads to a weight balanced matrix, commonly used in many multiagent systems.

Theorem 1: If Assumptions 1 and 2 hold, then $\rho((I-\Theta)W) < 1$ and the system (1) converges to the equilibrium point $\mathbf{x}^* = P_W \mathbf{x}(0)$ where

$$P_W = (I - (I-\Theta)W)^{-1}\Theta, \tag{5}$$

with $P_W \mathbb{1} = \mathbb{1}$. Furthermore, $\rho(P_W) = 1$ is a simple, positive, and strictly dominant eigenvalue and the left eigenvector of P_W associated to the eigenvalue 1 is $\Theta(I-\Theta)^{-1}\mathbb{1}$. Lastly, when $0 < \theta_i < 1$ for any i, then P_W is ES.

The proof is in Appendix B.

Remark 3: One of the properties of a FJ model on $\mathcal{G}$ nonnegative is that $\mathbf{x}^*$ belongs to the convex hull of the initial conditions: $\mathbf{x}_i^* \in \mathrm{co}(\mathbf{x}(0)) = [\min_i \mathbf{x}_i(0), \max_i \mathbf{x}_i(0)]$. As will be shown in the examples of Section VI (in particular, see Fig. 3), this is no longer true for SFJ models, even when convergence is guaranteed, i.e., when $\rho((I-\Theta)W) < 1$. For instance, in Fig. 3 Agent 6 leaves the convex hull.

The conditions of Theorem 1 are only sufficient but valid for all values of stubbornness obeying Assumption 1. If we restrict to stubbornness coefficients that are limited in a specific range, then the class of W for which the SFJ model is converging is significantly broader, and may include matrices whose $\omega(W)$ is not fully contained in the unit disc [see Fig. 1(c)].

Corollary 1: For a DT SFJ model (1) with W an ES matrix with $r(W) = \alpha > 1$, a sufficient condition for convergence, i.e. for $\rho((I-\Theta)W) < 1$, is to select Θ that ensures $r((I-\Theta)W) < 1$, i.e., $|\beta\varrho| < 1$ for every $\beta \in \omega(I-\Theta)$ and $\varrho \in \omega(W)$. The condition can be satisfied by setting all the stubbornness parameters sufficiently large, $\theta_i \in (\varepsilon, 1)$ for any i, where $\varepsilon = 1 - 1/\alpha$.

The proof is in Appendix B.

Remark 4: In some cases the model is robust to parameter uncertainty. For instance, it follows from Corollary 1 that, for all agents, choosing any value of stubbornness θ_i in the interval $(\varepsilon, 1)$ will not alter the stability of the model. Checking robustness simultaneously in Θ and W is instead much more difficult.

Remark 5: Having higher stubbornness values might appear counter-intuitive when the aim is to reach an equilibrium state, but it actually ensures that opinion values stay relatively close to the individuals' initial opinions and favors the stability of the SFJ model (1).

If we apply a more conservative condition that the stubbornness values are same for all agents, i.e., $\Theta = \varepsilon I$ with $0 < \varepsilon < 1$ (which we will use in Section V-C), then we get the following corollary.

Corollary 2: If Assumption 2 holds and $\Theta = \varepsilon I$ with $0 < \varepsilon < 1$, then P_W is ES and a radial matrix.

The proof is in Appendix B. When $r(W) > 1$, the convergence of the SFJ model, i.e., $\rho((I-\Theta)W) < 1$, does not guarantee $\rho(P_W) \leq 1$. Obviously, when $\rho(P_W) > 1$ then P_W cannot be a marginally stable matrix. For W ES with stubbornness values Θ such that $\rho((I-\Theta)W) < 1$, but W not satisfying the assumptions of Theorem 1, we can verify that $\rho(P_W) = 1$ through an eigenvalue localization approach. The following setup leads to a proposition equivalent to [21, Lemma 4], but for $\Theta \geq 0$. Let us assume that the first s agents are stubborn, while the remaining (subindex u) are not. Then, the dynamics can be partitioned as follows:

$$\begin{bmatrix}\mathbf{x}_s(t+1)\\ \mathbf{x}_u(t+1)\end{bmatrix} = \begin{bmatrix} I_s - \Theta_s & 0\\ 0 & I_u\end{bmatrix}\begin{bmatrix} W_{ss} & W_{su}\\ W_{us} & W_{uu}\end{bmatrix}\begin{bmatrix}\mathbf{x}_s(t)\\ \mathbf{x}_u(t)\end{bmatrix} + \begin{bmatrix}\Theta_s & 0\\ 0 & 0\end{bmatrix}\begin{bmatrix}\mathbf{x}_s(0)\\ \mathbf{x}_u(0)\end{bmatrix},$$

where $\mathbf{x}_s$ represent opinions of stubborn agents and $\mathbf{x}_u$ represent opinions of agents with zero stubbornness. Then, if we assume that the equilibrium exists, we can write the steady state values as

$$\mathbf{x}_u^* = (I_u - W_{uu})^{-1} W_{us}\mathbf{x}_s^*,$$

$$\mathbf{x}_s^* = \Big(I_s - (I_s - \Theta_s)\left(W_{ss} + W_{su}\left(I_u - W_{uu}\right)^{-1} W_{us}\right)\Big)^{-1} \Theta_s \mathbf{x}_s(0).$$

Denoting $\overline{W} = W_{ss} + W_{su}(I_u - W_{uu})^{-1}W_{us}$, we get

$$\mathbf{x}_s^* = \left(I_s - (I_s - \Theta_s)\,\overline{W}\right)^{-1}\Theta_s \mathbf{x}_s(0).$$

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With

$$P_{\overline{W}} = \left(I_s - (I_s - \Theta_s)\,\overline{W}\right)^{-1}\Theta_s, \tag{6}$$

we get $\mathbf{x}_s^* = P_{\overline{W}}\mathbf{x}_s(0)$. Notice that $\rho(\overline{W})$ need not be 1 and $\rho((I_s - \Theta_s)\overline{W})$ need not be less than 1; we only require $\lambda((I_s - \Theta_s)\overline{W}) \neq 1$. Since $\mathbf{x}_u^*$ depends on $\mathbf{x}_s^*$, we only need to prove that $\rho(P_{\overline{W}}) = 1$. We have $\overline{W}\mathbb{1}_s = \mathbb{1}_s$ which can be proved as follows:

$$\overline{W}\mathbb{1}_s = W_{ss}\mathbb{1}_s + W_{su}\left(I_u - W_{uu}\right)^{-1} W_{us}\mathbb{1}_s,$$

substituting $W_{ss}\mathbb{1}_s = \mathbb{1}_s - W_{su}\mathbb{1}_u$ and $W_{us}\mathbb{1}_s = (I_u - W_{uu})\mathbb{1}_u$ leads to

$$\overline{W}\mathbb{1}_s = \mathbb{1}_s - W_{su}\mathbb{1}_u + W_{su}\left(I_u - W_{uu}\right)^{-1}\left(I_u - W_{uu}\right)\mathbb{1}_u,$$

which proves $\overline{W}\mathbb{1}_s = \mathbb{1}_s$ and

$$P_W = \begin{bmatrix} P_{\overline{W}} & \mathbf{0} \\ \left(I_u - W_{uu}\right)^{-1} W_{us}P_{\overline{W}} & \mathbf{0} \end{bmatrix}.$$

Proposition 1: Consider an ES matrix W and stubbornness values Θ such that Assumption 1 is obeyed and $\rho((I-\Theta)W) < 1$. The condition $\rho(P_{\overline{W}}) \leq 1$ implies $\rho(P_W) \leq 1$. Whenever $(I_s - (I_s - \Theta_s)\overline{W})^{-1}$ and $(I_u - W_{uu})^{-1}$ exist, $\rho(P_{\overline{W}}) \leq 1$ is equivalent to $(I_s - \Theta_s^{-1})(I_s - \overline{W})$ having all eigenvalues outside (or on the boundary of) the unit disk centered at 1

$$\left|\lambda\left(I_s - (I_s - \Theta_s^{-1})(I - \overline{W})\right)\right| \geq 1.$$

B. Continuous Time Case

The results of the CT case are similar to those of the DT case: the DT SFJ model is in fact a discretized version of the CT SFJ model, and the DT adjacency matrix W can be derived from the Laplacian matrix L using $W = I - \phi L$, with proper selection of ϕ. One notable difference is that there is no "periodicity-2" complication in the sharp point property.

Assumption 3: $-L$ is an EEP matrix with $0 \in \omega(L)$ a sharp point.

Theorem 2: Under Assumptions 1 and 3, $-((I-\Theta)L + \Theta)$ is Hurwitz and the system (3) converges to the equilibrium point $\mathbf{x}^* = P_L\mathbf{x}(0)$ where

$$P_L = (\Theta + (I-\Theta)L)^{-1}\Theta, \tag{7}$$

with $P_L\mathbb{1} = \mathbb{1}$. Furthermore, $\rho(P_L) = 1$ is a simple, positive, and strictly dominant eigenvalue and the left eigenvector of P_L associated to eigenvalue 1 is $\Theta(I-\Theta)^{-1}\mathbb{1}$. Lastly, when $0 < \theta_i < 1$ for any i, then P_L is ES.

The proof is in Appendix B. Similarly to Theorem 1, the conditions of Theorem 2 are only sufficient but valid for all values of stubbornness obeying Assumption 1. By restricting the range of the stubbornness coefficients, the class of L for which the SFJ model is converging can be significantly expanded.

Corollary 3: For a CT SFJ model (1) with $-L$ an EEP matrix with $\mu(-H(L)) = \alpha > 0$, a sufficient condition for convergence, i.e., for $-\Theta - (I-\Theta)L$ Hurwitz, is to select Θ that ensures

$$\mu(-\Theta - (I-\Theta)^{\frac{1}{2}}H(L)(I-\Theta)^{\frac{1}{2}}) < 0.$$

The condition can be satisfied by setting the stubbornness parameters sufficiently large, namely, $\theta_i \in (\varepsilon, 1)$ for any i, where $\varepsilon = \alpha/(1+\alpha)$.

The proof is in Appendix B. When $\mu(-H(L)) > 0$, the convergence of the SFJ model, i.e., $-\Theta - (I-\Theta)L$ Hurwitz, does not guarantee $\rho(P_L) \leq 1$. More importantly, $\rho(P_L) > 1$ implies that P_L cannot be a marginally stable matrix. Similarly to Corollary 2 in DT, if the stubbornness values are the same for all agents, i.e., $\Theta = \varepsilon I$ with $0 < \varepsilon < 1$, then we get the following corollary.

Corollary 4: If Assumption 3 holds and $\Theta = \varepsilon I$ with $0 < \varepsilon < 1$, then P_L is ES and a radial matrix.

The proof is in Appendix B. For $-L$ EEP with stubbornness values Θ such that $-(I-\Theta)L - \Theta$ is Hurwitz, but $-L$ is not satisfying the assumptions of Theorem 2, the condition $\rho(P_L) = 1$ can be verified through an eigenvalue localization approach. Let us assume that the first s agents are stubborn and the remaining are not stubborn. The dynamics can be partitioned as follows:

$$\begin{bmatrix}\dot{\mathbf{x}}_s(t)\\ \dot{\mathbf{x}}_u(t)\end{bmatrix} = -\left(\begin{bmatrix} I_s - \Theta_s & 0 \\ 0 & I_u\end{bmatrix}\begin{bmatrix} L_{ss} & L_{su} \\ L_{us} & L_{uu}\end{bmatrix} + \begin{bmatrix}\Theta_s & 0\\ 0 & 0\end{bmatrix}\right)$$
$$\begin{bmatrix}\mathbf{x}_s(t)\\ \mathbf{x}_u(t)\end{bmatrix} + \begin{bmatrix}\Theta_s & 0\\ 0 & 0\end{bmatrix}\begin{bmatrix}\mathbf{x}_s(0)\\ \mathbf{x}_u(0)\end{bmatrix},$$

where $\mathbf{x}_s$ represent opinions of stubborn agents and $\mathbf{x}_u$ represent opinions of agent with zero stubbornness. Then, if we assume that the equilibrium exists, the final values can be written as

$$\mathbf{x}_u^* = -L_{uu}^{-1}L_{us}\mathbf{x}_s^*$$
$$\mathbf{x}_s^* = \left(\Theta_s + (I_s - \Theta_s)\left(L_{ss} - L_{su}L_{uu}^{-1}L_{us}\right)\right)^{-1}\Theta_s\mathbf{x}_s(0).$$

Denoting $\overline{L} = L_{ss} - L_{su}L_{uu}^{-1}L_{us}$, we get

$$\mathbf{x}_s^* = \left(\Theta_s + (I_s - \Theta_s)\,\overline{L}\right)^{-1}\Theta_s\mathbf{x}_s(0).$$

With

$$P_{\overline{L}} = \left(\Theta_s + (I_s - \Theta_s)\,\overline{L}\right)^{-1}\Theta_s, \tag{8}$$

we get $\mathbf{x}_s^* = P_{\overline{L}}\mathbf{x}_s(0)$. Since $\mathbf{x}_u^*$ depends on $\mathbf{x}_s^*$, we only need to prove that $\rho(P_{\overline{L}}) = 1$ where we have $\overline{L}\mathbb{1}_s = \mathbf{0}$, which can be proved as follows:

$$\overline{L}\mathbb{1}_s = L_{ss}\mathbb{1}_s - L_{su}L_{uu}^{-1}L_{us}\mathbb{1}_s,$$

after substituting $L_{ss}\mathbb{1}_s = -L_{su}\mathbb{1}_u$, and $L_{us}\mathbb{1}_s = -L_{uu}\mathbb{1}_u$, we get

$$\overline{L}\mathbb{1}_s = -L_{su}\mathbb{1}_u + L_{su}L_{uu}^{-1}L_{uu}\mathbb{1}_u = \mathbf{0},$$

and

$$P_L = \begin{bmatrix} P_{\overline{L}} & 0 \\ L_{uu}^{-1}L_{us}P_{\overline{L}} & 0\end{bmatrix}.$$

Proposition 2: Consider $-L$ EEP and Θ such that Assumption 1 is obeyed and $-\Theta - (I-\Theta)L$ is Hurwitz. The condition $\rho(P_{\overline{L}}) \leq 1$ implies $\rho(P_L) \leq 1$. Whenever $(\Theta_s + (I_s - \Theta_s)\overline{L})^{-1}$ and L_{uu}^{-1} exist, $\rho(P_{\overline{L}}) \leq 1$ is equivalent to $(I_s -$

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TABLE I
MAIN FEATURES OF THE SINGLE SFJ MODEL: SUMMARY

Case	Condition	Behavior	Theorems
1	DT: W is ES with a strictly dominant sharp point at $1 \in \omega(W)$; CT: the Laplacian matrix $-L$ is EEP with a sharp point at $0 \in \omega(L)$	The SFJ model has a globally asymptotically stable equilibrium point for all stubbornness values. The opinions may/may not converge inside the convex hull of the initial conditions.	Theorems 1 and 2
2	DT: W is ES with $r(W) > 1$; CT: $-L$ is EEP with $\mu(-H(L)) > 0$	The SFJ model may have a globally asymptotically stable equilibrium point or be unstable, depending on the stubbornness values. The opinions may/may not converge inside the convex hull of the initial conditions.	Corollary 1 and 3
3	DT: W is not ES; CT: $-L$ is not EEP	The model may be (and often is) unstable.	$--$

TABLE II
MAIN FEATURES OF THE CONCATENATED SFJ MODEL: SUMMARY

Case	Condition	Behavior	Theorems
1	DT: W is ES with a strictly dominant sharp point at $1 \in \omega(W)$; CT: the Laplacian matrix $-L$ is EEP with a sharp point at $0 \in \omega(L)$	The concatenated SFJ model converges to a consensus point inside the convex hull. The opinions may leave the convex hull transiently.	Theorem 3
2	DT: W is ES with $r(W) > 1$; CT: $-L$ is EEP with $\mu(-H(L)) > 0$	The concatenated SFJ model may or may not have a stable response even when the single SFJ model converges to an equilibrium point (if outside the convex hull).	$--$
3	DT: W is not ES; CT: $-L$ is not EEP	The concatenated SFJ model typically diverges.	$--$

$\Theta_s^{-1})\overline{L}$ having all eigenvalues outside (or on the boundary of) the unit disk centered at 1

$$\left|\lambda\left(I_s - (I_s - \Theta_s^{-1})\overline{L}\right)\right| \geq 1.$$

The proof is in Appendix B.

C. Single SFJ Model: Summary

For the SFJ models (1) and (3), the main stability conditions we have obtained are summarized in Table I.

A few issues are worth emphasizing.

Remark 6: The DT/CT cases are not completely specular: in DT we have to add the extra property that the sharp point at $1 \in \omega(W)$ is strictly dominant, condition which is absent in CT. In this respect, the CT SFJ model is slightly more general. The situation is analogous to what happens for nonnegative matrices: in DT we must add the condition that the adjacency matrix is aperiodic. The analogy between strict dominance of the sharp point for signed matrices and aperiodicity of nonnegative matrices has in our knowledge not been observed before in the literature.

Remark 7: The properties of ES, EEP, and sharp points are global characteristics of the adjacency matrix of the interaction graph and cannot be deduced by looking at one agent and its local interaction pattern. This reflects the fact that unlike the unsigned FJ model, where stability is always assured, the SFJ model can exhibit instability. Consequently, assessing stability in SFJ dynamics necessitates understanding the collective dynamics and global interaction pattern among all the agents in the network. This reflects more properly what happens in real opinion dynamics, where the single-agent perspective is inadequate for understanding the overall emerging dynamics.

V. EXTENSIONS

A. Concatenated SFJ Models

A concatenated SFJ model is a two-time scale model representing a sequence of discussion events, each of which is represented by a SFJ model (DT or CT). Using the setting we developed in [11], a second index $s = 1, 2, \ldots$ provides the clock for the sequence of discussion events, hence the opinion vector becomes $\mathbf{x}(s, t)$, and the SFJ dynamics at event s is

$$\mathbf{x}(s, \infty) = P\,\mathbf{x}(s, 0), \tag{9}$$

where $P \in \{P_W, P_L\}$ depending on whether we are dealing with DT or CT dynamics. Concatenation refers to the endpoint of the $s-1$ meeting becoming the initial condition of the s meeting

$$\mathbf{x}(s, 0) = \mathbf{x}(s-1, \infty),$$

with the understanding that $\mathbf{x}(0, \infty) = \mathbf{x}(1, 0)$ is the initial opinion of the agents at the beginning of the first meeting. This implies that (9) can be rewritten as

$$\mathbf{x}(s, \infty) = P\,\mathbf{x}(s-1, \infty), \quad P \in \{P_W, P_L\} \tag{10}$$

i.e., a DT dynamics in s in which P is a signed matrix solving an SFJ model. Notice that for simplicity in this section we assume that P is the same on each discussion event. Under the assumptions of Theorems 1 and 2, P has special properties, and we have that the concatenated SFJ system (10) converges to consensus.

Theorem 3: Under Assumption 1, if

1) [DT case:] Assumption 2 holds,
2) [CT case:] Assumption 3 holds,

then the concatenated SFJ model (10) converges to consensus.

The proof is in Appendix C. As discussed for the DT case, the violation of convex hull is linked to negative terms in matrix P (P_W in DT, P_L in CT), but the likelihood of violation increases with the non-normality of the matrix P. We observe that starting from W or L normal is not enough to guarantee that P_L/P_W is normal. The following Lemma (whose proof is in Appendix C) provides a sufficient condition for the matrix P to be normal.

Lemma 12: The matrix P in (10) is normal if $\Theta = \varepsilon I$, where $0 \leq \varepsilon \leq 1$ and

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1) [DT Case:] W is ES and normal.
2) [CT Case:] $-L$ is EEP and normal.

B. Bipartite Signed Graphs and Bipartite Consensus

Consider a signed digraph $\mathcal{G}$ with a "bipartite structure," that is, such that the DT weighted interaction matrix, call it W_b, is SES or, in CT, the signed Laplacian $-L_b$ is SEEP. Then, we know from Definition 4 that there exists a diagonal signature matrix $\Omega = \mathrm{diag}(\sigma)$ such that $W_b = \Omega W \Omega$ with W ES and, similarly, $L_b = \Omega L \Omega$ with $-L$ EEP. Consider the associated SFJ models

$$\text{DT:} \quad \mathbf{x}(t+1) = (I-\Theta)\, W_b + \Theta \mathbf{x}(0), \tag{11}$$

$$\text{CT:} \quad \dot{\mathbf{x}}(t) = -((I-\Theta)\, L_b + \Theta)\mathbf{x}(t) + \Theta \mathbf{x}(0). \tag{12}$$

It follows by construction that these systems are gauge transformations of the SFJ models (1) and (3) we considered in the previous sections, obtained by simply applying a change of basis $\mathbf{z}(t) = \Omega \mathbf{x}(t)$.

The results of Theorems 1 and 2 can then be generalized to more complex interaction graphs, in which $\mathcal{G}$ has a much larger quantity of antagonistic interactions than for the ES (or EEP) case. The following theorems are given without proof, since they can be straightforwardly deduced from Theorems 1 and 2 modulo the gauge transformation. First, we show that the numerical range remains the same after gauge transformation.

Lemma 13: If $W_b = \Omega W \Omega$ and $L_b = \Omega L \Omega$ with $\Omega = \mathrm{diag}(\sigma)$, and if

1) [DT Case:] Assumption 2 holds, then W_b is SES with $1 \in \omega(W_b)$ a strictly dominant sharp point.
2) [CT Case:] Assumption 3 holds, then $-L_b$ is SEEP with $0 \in \omega(L_b)$ a sharp point.

The proof is straightforward using Lemma 7.

Assumption 4: W_b is a SES matrix with $1 \in \omega(W_b)$ a strictly dominant sharp point.

Assumption 5: $-L_b$ is a SEEP matrix with $0 \in \omega(L_b)$ a sharp point.

Theorem 4: Under Assumption 1, if

1) [DT case:] Assumption 4 holds, then $\rho((I-\Theta)W_b) < 1$ and the SFJ model (11) converges to $\mathbf{x}^* = P_{W_b}\mathbf{x}(0)$ where
$$P_{W_b} = (I-(I-\Theta)W_b)^{-1}\,\Theta \tag{13}$$
is a marginally stable matrix, with $P_{W_b}\boldsymbol{\sigma} = \boldsymbol{\sigma}$.
2) [CT case:] Assumption 5 holds, then $-((I-\Theta)L_b + \Theta)$ is Hurwitz, and the SFJ model (12) converges to $\mathbf{x}^* = P_{L_b}\mathbf{x}(0)$ where
$$P_{L_b} = (\Theta + (I-\Theta)L_b)^{-1}\Theta \tag{14}$$
is a marginally stable matrix, with $P_{L_b}\boldsymbol{\sigma} = \boldsymbol{\sigma}$.

Concatenated SFJ models can be associated to (11) and (12)

$$\mathbf{x}(s,\,\infty) = P_b\,\mathbf{x}(s-1,\infty), \quad P_b \in \{P_{W_b},\, P_{L_b}\}. \tag{15}$$

Theorem 3 generalizes straightforwardly to the following one, which however leads to bipartite consensus rather than to consensus.

Theorem 5: Under Assumption 1, if

1) [DT case:] Assumption 4 holds,
2) [CT case:] Assumption 5 holds,

then the concatenated SFJ model (15) converges to bipartite consensus, with bipartition given by the signature of $\boldsymbol{\sigma}$.

C. Time-Varying SFJ Concatenations

So far, in our work on the concatenated SFJ model we have assumed that the interaction pattern and stubbornness values for the individuals remain the same over different discussion events. In practical scenarios, however, the interaction between individuals and the attachment of individuals to their initial opinions may change over different discussion events. For the time-varying concatenated SFJ case, we have then

$$\mathbf{x}(s,\,\infty) = P(s)\,\mathbf{x}(s-1,\infty), \;\; P(s) \in \{P_W(s),\, P_L(s)\} \tag{16}$$

where

$$P_W(s) = (I-(I-\Theta(s))W(s))^{-1}\,\Theta(s), \tag{17}$$

$$P_L(s) = (\Theta(s) + (I-\Theta(s))L(s))^{-1}\,\Theta(s), \tag{18}$$

and both $\Theta(s)$ and $W(s)$ (or $L(s)$) depend on the index s. According to [20, Thm. 8], if $P(s)$ is ES and normal for all s, the system (16) achieves consensus. Assuming $\Theta(s) = \varepsilon(s)I$ with $0 < \varepsilon(s) < 1$, $W(s)$ ES and normal in DT, and $-L(s)$ EEP and normal in CT, then, from Lemma 12, we have that $P(s)$ is ES and normal.

Similarly to the time-invariant case, we can extend the class of interaction matrices from normal to those having a sharp point at 1 for $\omega(W(s))$ in DT and at 0 for $\omega(L(s))$ in CT. The following assumptions replace Assumptions 2 and 3.

Assumption 6: $W(s)$ is an ES matrix with $1 \in \omega(W(s))$ a strictly dominant sharp point.

Assumption 7: $-L(s)$ is an EEP matrix with $0 \in \omega(L(s))$ a sharp point.

Theorem 6: Assuming that $\Theta(s) = \varepsilon(s)I$ with $0 < \varepsilon(s) < 1$, if $\forall\, s$

1) [DT case:] $W(s)$ is ES with $1 \in \omega(W(s))$ a sharp point,
2) [CT case:] Assumption 7 holds,

then the time-varying concatenated SFJ model (16) converges to consensus.

The proof is in Appendix C. Notice that in Theorem 6 we do not require the sharp point to be strictly dominant. This follows from the radiality of $P(s)$, but it is only valid when the stubbornness coefficients at each event are all equal (to $\varepsilon(s)$). The all-equal stubbornness condition is however restrictive. A more practical case of different stubbornness values among agents can also be treated, at the cost of imposing strict dominance of the sharp point on the matrix W. When the stubbornness values are unequal across agents but constant over s, we have the following.

Theorem 7: Under Assumption 1 and assuming that $\Theta(s) = \Theta \succ 0$ is fixed over different discussions, if $\forall\, s$

1) [DT case:] Assumption 6 holds,
2) [CT case:] Assumption 7 holds,

then the time-varying concatenated SFJ model (16) converges to consensus.

The proof is in Appendix C. As its proof shows, Theorem 7 is based on the construction of a quadratic CLF for the time-

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varying system (16). Its generating matrix, $\Theta(I-\Theta)^{-1}$, is known because of the special choice made for the stubbornness values, which are kept constant through the events. When both $\Theta(s)$ and $W(s)$ (or $L(s)$) change with s, a similar construction may or may not exist, depending on the stubbornness values. The following theorem (whose proof is in Appendix C) says that for arbitrary $\Theta(s)$, the concatenated SFJ converges if a radially unbounded CLF can be found for all possible discussion events. Consider a matrix $\Gamma \in \mathbb{R}^{(n-1)\times n}$ whose rows form an orthonormal basis for $\text{span}(\mathbb{1})^{\perp}$, i.e., $\Gamma\Gamma^T = I_{n-1}$, $\Gamma\mathbb{1} = \mathbf{0}$.

Theorem 8: Under Assumption 1, if $\forall\, s$

1) [DT case:] Assumption 6 holds,
2) [CT case:] Assumption 7 holds,

and $\exists\, Q = Q^T \succ 0$ such that $\forall s$

$$\Gamma P^T(s) Q P(s) \Gamma^T - \Gamma Q \Gamma^T \prec 0,\ P(s) \in \{P_W(s), P_L(s)\} \tag{19}$$

is satisfied, then $V(\mathbf{x}) = \mathbf{x}^T Q \mathbf{x}$ is a CLF and the time-varying concatenated SFJ model (16) converges to consensus.

Remark 8: Theorem 8 requires finding a common quadratic Lyapunov function for all possible discussion events. This amount to formulating (19) as a set of LMIs to be solved simultaneously, one for each $P(s) \in \{P_W(s), P_L(s)\}$. The problem may or may not have a solution, depending on the $P(s)$.

VI. Examples

Consider a social network consisting of six individuals influencing each other according to the interaction graph depicted in Fig. 2. For this graph, we can use either DT or CT SFJ models. In the DT model, we can set $w_{ij} = a_{ij}$ and $w_{ii} = 1 - \sum_j a_{ij}$, while for the CT model we can derive the Laplacian matrix using $L = \Sigma_{\text{in}} - A$. The graph is signed but not structurally balanced, and just by looking at the (signed) topology in Fig. 2 it is impossible to deduce the outcome of the SFJ dynamics, which will depend on the specific values chosen for the edge weights a_{ij} and possibly from the stubbornness coefficients θ_i.

For this example, we study below several possible cases for what concerns the stability behavior of the SFJ and concatenated SFJ models. Let us call

$$\mathbf{a}_i = \{a_{12}, a_{21}, a_{23}, a_{32}, a_{34}, a_{43}, a_{36}, a_{63}, a_{45}, a_{54}, a_{46}, a_{64}\}$$

the tuple of interaction weights from Fig. 2. The index i denotes different numerical choices of the weights.

1) Choosing

$$\mathbf{a}_1 = \{0.6,\ 0.6,\ 0.1,\ 0.1,\ -0.35,\ 0.6,\ 0.6,\ -0.35,\ 0.1,\ 0.1,\ -0.35,\ 0.6\},$$

we have that the SFJ model belongs to *Case 1* of Table I. Consequently, Theorems 1 and 2 can be applied: any stubbornness value results in a stable DT/CT SFJ model and a concatenated SFJ model converging to consensus. Violation of the convex hull invariance instead depends on the choice of the θ_i. For instance in the DT case, with stubbornness values $\Theta = \text{diag}([0.74,\ 0.8,\ 0.95,\ 0.95,\ 0.79,\ 0.55])$, we get that the stable DT SFJ model escapes the convex hull (see Fig. 3) and, in the concatenated case, exhibits a transient amplification followed by convergence [see Fig. 4(a)]. For instance, the negative edge between individuals 3 and 6 makes them repel each other on the shorter time scale, but it is not enough to keep them apart on the longer time scale of the sequence of discussions. This transient amplification is a well known phenomena which follows from the nonnormality of matrix P_W [25]. In our case, we have $r(P_W) > \rho(P_W) = 1$ and the difference between numerical radius and spectral radius, i.e., $r(P_W) - \rho(P_W)$, is a measure of nonnormality, see [25].

2) Choosing instead

$$\mathbf{a}_2 = \{0.19,\ 0.31,\ 0.39,\ 0.92,\ -0.42,\ 0.94,\ 0.18,\ -0.57,\ 0.81,\ 0.08,\ -0.99,\ 0.39\},$$

we have *Case 2* of Table I. In this case stability depends on the choice of stubbornness. According to Corollaries 1 and 3, if $\theta_i > 0.3014$ for any i, the DT and CT SFJ model will be stable. The constraint $\theta_i > 0.3014$, for any i, renders $r((I-\Theta)W) < 1$ and provides a condition which is sufficient but not necessary for $\rho((I-\Theta)W) < 1$. For instance $\Theta = 0.1I$ implies stability, while $\Theta = \text{diag}([0.4,\ 0.2,\ 0.9,\ 0.9,\ 0.4,\ 0.1])$ implies instability. Lastly, as $r(W) > 1$, the convergence of the single SFJ model, i.e., $\rho((I-\Theta)W) < 1$, does not guarantee $\rho(P_W) \leq 1$. For instance, with $\Theta = \text{diag}([0.74,\ 0.8,\ 0.99,\ 0.99,\ 0.79,\ 0.55])$, convergence is achieved, i.e, $\rho((I-\Theta)W) < 1$, but $\rho(P_W) > 1$, meaning that the concatenated SFJ model keeps expanding and asymptotically diverges. In particular, we can notice in Fig. 4(b) that the opinion of agent 6 always opposes the opinion of agent 3, meaning that a form of polarization is created by the model, in spite of the lack of structural balance in the graph.

The same interaction graph of Fig. 2 (but with possibly a different sign pattern) can be used to investigate the behavior of the bipartite signed graph of Section V-B and of the time-varying concatenated case of Section V-C.

3) Choosing edges weights with a different sign pattern

$$\mathbf{a}_3 = \{-0.6,\ -0.6,\ -0.2,\ -0.2,\ 0.1,\ -0.14,\ 0.14,\ -0.1,\ 0.2,\ 0.2,\ 0.1,\ -0.14\},$$

we have that W is SES with $1 \in \omega(W)$ a strictly dominant sharp point and gauge matrix $\Omega = \text{diag}([1,\ -1,\ 1,\ -1,\ -1,\ 1])$. Theorem 5 applies, and the gauge transformed adjacency matrix $\Omega W \Omega$ results in the sign pattern of Fig. 2. For $\Theta = \text{diag}([0.6,\ 0.6,\ 0.6,\ 0.2,\ 0.3,\ 0.5])$, the opinions propagate to bipartite consensus as shown in Fig. 4(c).

4) Consider a time varying case in which the graph switches between the tuples of parameters $\mathbf{a}_1$ and

$$\mathbf{a}_4 = \{0.6,\ 0.6,\ 0.1,\ 0.1,\ 0.5,\ -0.3,\ -0.3,\ 0.5,\ 0.1,\ 0.1,\ 0.5,\ -0.3\}.$$

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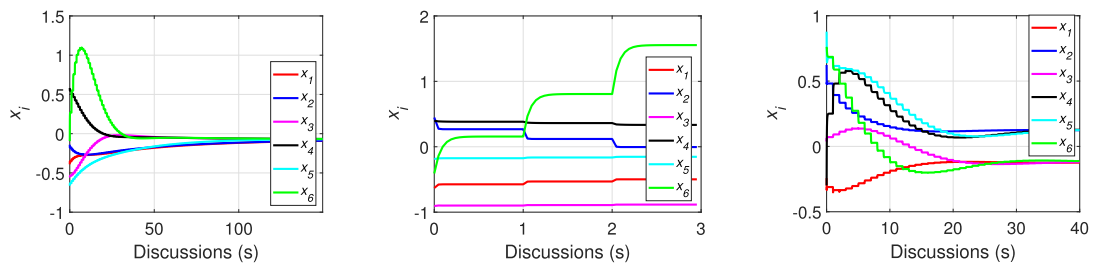

(a) (b) (c)

Fig. 4. Propagation of opinions in the concatenated SFJ model of Fig. 2.

For both tuples, W is ES with $1 \in \omega(W(s))$ is a strictly dominant sharp point. The time-varying DeGroot model obtained when $\Theta(1) = \Theta(2) = \mathbf{0}$ is unstable, as $\rho(W(2)W(1)) > 1$. But with varying stubbornness values $\Theta(1) = \mathrm{diag}([0.65, 0.38, 0.92, 0.21, 0.54, 0.51])$ and $\Theta(2) = \mathrm{diag}([0.17, 0.02, 0.81, 0.72, 0.87, 0.71])$, in the concatenated SFJ model (16) we achieve consensus, as $\rho(P_W(2)P_W(1)) = 1$. Theorem 8 applies, with quadratic CLF obtained from the matrix

$$Q=\begin{bmatrix} 1.49 & -0.19 & -0.70 & -0.04 & -0.32 & -0.22 \\ -0.19 & 0.76 & -0.34 & -0.05 & -0.07 & -0.10 \\ -0.70 & -0.34 & 2.8 & -0.40 & -0.89 & -0.47 \\ -0.04 & -0.05 & -0.40 & 1.09 & -0.32 & -0.29 \\ -0.32 & -0.07 & -0.89 & -0.32 & 1.91 & -0.31 \\ -0.22 & -0.10 & -0.47 & -0.29 & -0.31 & 1.39 \end{bmatrix}.$$

5) When the stubbornness coefficients are kept constant over all discussion issues, then the construction of a CLF simplifies: considering a graph that switches between $\mathbf{a}_1$ and $\mathbf{a}_4$ and choosing constant stubbornness values $\Theta = \mathrm{diag}([0.21, 0.5, 0.89, 0.06, 0.42, 0.05])$ in all discussions, we still get $\rho(P_W(2)P_W(1)) = 1$, with quadratic CLF matrix $Q = (I - \Theta)^{-1}\Theta$. This is the case treated in Theorem 7.

When the stubbornness coefficient change over the events and Theorem 7 no longer holds, the system might not reach a consensus, not even when the topology is kept fixed and only the stubbornness changes.

6) A time-varying case, in which the edge weights are fixed to $\mathbf{a}_1$ but the stubbornness values switch between $\Theta(1) = \mathrm{diag}([0.86, 0.21, 0.99, 0.82, 0.40, 0.04])$ and $\Theta(2) = \mathrm{diag}([0.18, 0.38, 0.9, 0.04, 0.77, 0.98])$, results in $\rho(P_W(2)P_W(1)) = 1.15$.

When some stubbornness values are zero and the DT/CT SFJ model is stable, then the stability of the concatenated model can be found using the Propositions 1 and 2. See next example for the DT case.

7) Choosing $\mathbf{a}_2$ with $\Theta = \mathrm{diag}([0, 0, 0, 0.8, 0.1, 0.3])$, we get $\rho((I - \Theta)W) < 1$, i.e., the system (1) is stable. With

$$\overline{W} = \begin{bmatrix} 1.885 & 0.81 & -1.695 \\ 0.08 & 0.92 & 0 \\ -0.6075 & 0 & 1.6075 \end{bmatrix},$$

we have that $\lambda((I_s - \Theta_s^{-1})(I_s - \overline{W})) = 1.81$ is inside a unit disk centered at 1. Hence, Proposition 1 implies that the concatenated SFJ model is unstable and indeed we get $\rho(P_W) = 1.23 > 1$.

VII. Conclusion and Outlook

In this article, we have given sufficient conditions for existence of a stable equilibrium point in the SFJ model. If W is ES with a strictly dominant sharp point in $1 \in \omega(W)$ for the DT case (or if $-L$ EEP with a sharp point in $0 \in \omega(L)$ for the CT case), then a stable equilibrium point exists for all values of stubbornness. We also show that the same conditions are required for the convergence to consensus in the concatenated case, as well as for dealing with the bipartite consensus case for a more general class of concatenated signed graphs. Further generalizations of these sufficient conditions are possible, but only if we restrict to specific values of stubbornness. Lastly, we present several specific instances of time-varying concatenated SFJ models, where the interaction graph and/or stubbornness of agents vary between discussion events.

For what concerns the violation of the convex hull of the initial conditions, this feature can appear in several guises as follows:

1) at the end of a discussion for the single SFJ model;
2) during the transient in a concatenated SFJ model;
3) in all discussions of a concatenated SFJ model.

At the moment, no precise condition is available in any of the three cases. Clearly, a "nonnormality inducing" condition like $\rho(P_W) = 1$ and $r(P_W) > 1$ favors its appearance, but it is neither necessary nor sufficient. More research is needed to clarify the issue.

Appendix A
Preliminaries and Problem Formulation

Proof of Lemma 4: Consider the matrix $\overline{Q} = Q - I$ where Q is ES. This $\overline{Q}$ is an EEP matrix, as $\overline{Q} + I \overset{\vee}{>} 0$. When $\rho(Q) = 1$ is a simple, positive, and strictly dominant eigenvalue, it implies $\mu(Q - I) = 0$ is simple. ■

Proof of Lemma 7: This lemma follows from the unitary similarity invariance property (1.2.8) in [29] that for a unitary matrix U, $\omega(U^*QU) = \omega(Q)$. ■

Proof of Lemma 8: Consider the ES matrix W with the property $W\mathbb{1} = \mathbb{1}$, i.e., $1 \in \Lambda(W)$. Using Lemma 6, we get that the left and right eigenvector corresponding to the eigenvalue

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$\lambda(W) = 1$ are identical when $1 \in \omega(W)$ is a sharp point. This implies that $\mathbb{1}^T W = \mathbb{1}^T$, and therefore W is EDS. ■

Proof of Lemma 10: Consider the EEP matrix L with the property $L\mathbb{1} = \mathbf{0}$, i.e., $0 \in \Lambda(L)$. Using Lemma 6, we get that the left and right eigenvector corresponding to the eigenvalue $0 \in \Lambda(L)$ are identical when $0 \in \omega(L)$ is a sharp point. This implies that $\mathbb{1}^T L = \mathbf{0}^T$, and therefore L is weight balanced. ■

Appendix B
Convergence in SFJ Models

Proof of Theorem 1: To prove that $\rho((I-\Theta)W) < 1$ where W has a strictly dominant sharp point at $1 \in \omega(W)$, we first prove it for W that is a radial matrix.

The stability analysis of the affine system (1) and of the linear system obtained from (1) applying the change of basis $\mathbf{z}(t) = \mathbf{x}(t) - \mathbf{x}^*$ are the same whenever $\mathbf{x}^*$ exists unique, that is, when $\rho((I-\Theta)W) < 1$. Hence, this condition can be checked on the system $\mathbf{z}(t+1) = (I-\Theta)W\mathbf{z}(t)$.

Assume $\theta_i \in (0,\, 1)$ for some $i \in 1, \ldots, n$. Using the properties of two-norms and denoting $D = I - \Theta$, we get

$$\|DW\|_2 = \max_{\mathbf{x},\mathbf{y}\in\mathbb{S}^{n-1}} |\mathbf{x}^T DW\mathbf{y}| \leq \max_{\mathbf{x},\mathbf{y}\in\mathbb{S}^{n-1}} \|D\mathbf{x}\|_2 \|W\mathbf{y}\|_2 \leq 1$$

where the first inequality follows from the Cauchy–Schwarz inequality, which is met with equality *iff* $\kappa D\mathbf{x} = W\mathbf{y}$ where $\kappa \in \mathbb{R}$ (i.e., $W\mathbf{y}$ is in the span of $D\mathbf{x}$). The second inequality is met with equality *iff* $D\mathbf{x} = \mathbf{x}$, which implies that it must be $x_i = 0$, and $W\mathbf{y} = \mathbf{y}$, which implies that $\mathbf{y}$ must be a positive vector, because W is ES and a radial matrix, which implies that $\rho(W)I - W^T W \succeq 0$ and hence $\rho(W) = \|W\| = 1$. Therefore, the equality $\|DW\|_2 = 1$ is satisfied when the condition $\kappa D\mathbf{x} = W\mathbf{y} \Rightarrow \kappa\mathbf{x} = \mathbf{y}$ is met, which is impossible as $\mathbf{y}$ is a positive vector and $x_i = 0$. This proves that $\rho(DW) < 1$ under Assumption 1 when W is a radial matrix.

Since $DW = D^{\frac{1}{2}} D^{\frac{1}{2}} W D^{\frac{1}{2}} D^{-\frac{1}{2}}$, the matrices DW and $D^{\frac{1}{2}} W D^{\frac{1}{2}}$ are similar, with transformation matrix $D^{\frac{1}{2}}$. To prove that $\rho(DW) < 1$, it suffices then to demonstrate that $\rho(D^{\frac{1}{2}} W D^{\frac{1}{2}}) < 1$.

For an ES matrix W with $1 \in \omega(W)$ as a strictly dominant sharp point, we have that $H(W)$ is a radial matrix with $\mathbb{1}$ as both the left and right eigenvector associated with the eigenvalue $\rho(H(W)) = 1$. Then, from the initial part of the proof, $\mu(D^{\frac{1}{2}} H(W) D^{\frac{1}{2}}) < 1$.

For $D \succ 0$, we establish

$$\Lambda(D^{\frac{1}{2}} H(e^{i\vartheta} W) D^{\frac{1}{2}}) \subset \omega(D)\omega(H(e^{i\vartheta} W)).$$

With assumptions of the Theorem, we have $\mu(H(e^{i\vartheta} W)) < 1$ and $|\beta| \leq 1$ with $\beta \in \omega(D)$, which implies $\mu(D^{\frac{1}{2}} H(e^{i\vartheta} W) D^{\frac{1}{2}}) < 1$ for $0 < \vartheta < 2\pi$.

Consequently, with two assertions: 1) $\mu(D^{\frac{1}{2}} H(W) D^{\frac{1}{2}}) < 1$, and 2) $\mu(D^{\frac{1}{2}} H(e^{i\vartheta} W) D^{\frac{1}{2}}) < 1$ for $0 < \vartheta < 2\pi$, we deduce that $\mu(D^{\frac{1}{2}} H(e^{i\vartheta} W) D^{\frac{1}{2}}) < 1$ for all $\vartheta \in \mathbb{R}$, implying $r(D^{\frac{1}{2}} W D^{\frac{1}{2}}) < 1$ using Lemma 5. Finally, we conclude that $\rho(D^{\frac{1}{2}} W D^{\frac{1}{2}}) \leq r(D^{\frac{1}{2}} W D^{\frac{1}{2}}) < 1$.

The above implies $\mathbf{z}(t) \xrightarrow{t\to\infty} \mathbf{0}$, and $\mathbf{x}(t) \xrightarrow{t\to\infty} \mathbf{x}^*$. The value of $\mathbf{x}^*$ is computed from the solution of (1)

$$\mathbf{x}(t) = \left((DW)^t + \sum_{k=0}^{t-1} (DW)^k \Theta \right) \mathbf{x}(0).$$

When $t \to \infty$, the first term vanishes, and the second term becomes the following converging Neumann series

$$\mathbf{x}^* = \lim_{t\to\infty} \sum_{k=0}^{t-1} (DW)^k \Theta = (I - DW)^{-1} \Theta.$$

To show that P_W has eigenvalue 1, we can use the following chain of equalities:

$$\begin{aligned} DW\mathbb{1} &= \mathbb{1} - \Theta\mathbb{1} \\ (I - DW)\,\mathbb{1} &= \Theta\mathbb{1} \\ \mathbb{1} &= ((I - DW))^{-1}\,\Theta\mathbb{1} = P_W\mathbb{1}. \end{aligned}$$

We also need to show that $\rho(P_W) = 1$ is a simple, strictly dominant, positive eigenvalue and that the left eigenvector corresponding to $\rho(P_W) = 1$ is positive (for $\theta_i > 0$, for any i) or nonnegative (when $\theta_i = 0$ for some i).

From the initial part of the proof, we have $P_W\mathbb{1} = \mathbb{1}$ meaning that $\lambda(P_W) = 1$ is an eigenvalue of P_W. Furthermore, we have $r(DW) < 1$ which implies $\mathbf{x}^T(I - DW)\mathbf{x} > 0$. From this it follows that $\mathbf{x}^T(D(I-W) + \Theta)\mathbf{x} > 0$. This also implies $\mathbf{x}^T(D^{1/2}(I-W)D^{1/2} + \Theta)\mathbf{x} > 0$ through the transformation $T = D^{1/2}$. For W such that $1 \in \omega(W)$ is a sharp point, we have $\mathbf{x}^T(I-W)\mathbf{x} \geq 0$ and $\mathbf{x}^T D\mathbf{x} > 0$, leading to $\mathbf{x}^T(D^{1/2}(I-W)D^{1/2})\mathbf{x} \geq 0$. We prove the statement of the theorem by contradiction. Suppose $\rho(P_W) = \eta > 1$ with right eigenvector $\mathbf{v}$, then we obtain $(I - DW)^{-1}\Theta\mathbf{v} = \eta\mathbf{v}$, which can be rewritten as

$$(D(I-W) + \Theta((\eta-1)/\eta))\,\mathbf{v} = \mathbf{0}.$$

Consequently, this after transformation implies that

$$\mathbf{x}^T \left(D^{1/2}(I-W)D^{1/2} + \Theta((\eta-1)/\eta) \right) \mathbf{x} = 0.$$

The above equation contradicts

$$\mathbf{x}^T \left(D^{1/2}(I-W)D^{1/2} + \Theta \right) \mathbf{x} > 0.$$

In fact $1 > (\eta-1)/\eta > 0$, and both $D^{1/2}(I-W)D^{1/2} \succeq 0$ and $\Theta \succeq 0$. To prove that $\rho(P_W) = 1$ is simple, notice that for $\eta = 1$ and $\mathbf{v} = \mathbb{1}$, we have $D(I-W)\mathbf{v} = 0$ and $corank(D(I-W)) = 1$, as $\rho(W) = 1$ is simple.

Lastly, to prove that the left eigenvector of P_W is nonnegative, notice that from Lemma 8, $\mathbb{1}$ is the left eigenvector of W associated to the eigenvalue 1. Using the fact that $\rho(DW) = \rho(WD) < 1$ and that $(I - WD)^{-1}$ exists, we have

$$\begin{aligned} \mathbb{1}^T (I - WD) &= \mathbb{1}^T\Theta \\ \mathbb{1}^T\Theta\,(I - WD)^{-1} &= \mathbb{1}^T \\ \mathbb{1}^T\Theta\,(I - WD)^{-1} D^{-1}\Theta &= \mathbb{1}^T D^{-1}\Theta \\ \mathbb{1}^T D^{-1}\Theta\,(I - DW)^{-1}\,\Theta &= \mathbb{1}^T D^{-1}\Theta \end{aligned}$$

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which proves that $\Theta D^{-1}\mathbb{1}$ is the left eigenvector of P_W associated to the eigenvalue 1. With $\Theta \succeq 0$ and $D \succ 0$, we have $\Theta D^{-1}\mathbb{1} \geq 0$. This completes the proof. ■

Proof of Corollary 1: Letting $D = I - \Theta \succ 0$, $\beta \in \omega(D)$, $\varrho \in \omega(W)$, and utilizing the property $\Lambda(DW) \subset \beta\varrho$, alongside the observations that $|\beta| < 1/\alpha$ and $|\varrho| \leq \alpha$, we can conclude that $\rho(DW) < 1$. ■

Proof of Corollary 2: Using Theorem 1, P_W is ES for the given assumptions. To prove that P_W is a radial matrix, notice that $\omega(I - (1-\varepsilon)W)$ has a sharp point at ε, which results in a sharp point at ε^{-1} for $\omega((I-(1-\varepsilon)W)^{-1})$ and at 1 for $\omega(P_W)$ where $P_W = \varepsilon(I-(1-\varepsilon)W)^{-1}$. Also, notice that $0 \notin \omega(P_W)$ as $r((1-\varepsilon)W) < 1$, then using the property $\Lambda(P_W^T P_W) \subset \beta^2$ with $\beta \in \omega(P_W)$ we get that $\rho(P_W^T P_W) = 1$ is a simple, positive, and strictly dominant eigenvalue. As $P_W^T P_W$ is symmetric, P_W is a radial matrix. ■

Proof of Proposition 1: Under the assumptions of the Proposition, Θ_s is invertible and $(I_s - (I_s - \Theta_s)\overline{W})^{-1}$ exists, and we can write $P_{\overline{W}} = (\overline{W} + \Theta_s^{-1}(I_s - \overline{W}))^{-1}$. Hence, $\rho(P_{\overline{W}}) \leq 1$ is equivalent to $|\lambda((\overline{W} + \Theta_s^{-1}(I_s - \overline{W}))^{-1})| \leq 1$ for all eigenvalues of $(\overline{W} + \Theta^{-1}(I_s - \overline{W}))^{-1}$, which corresponds to say that for all eigenvalues of $\overline{W} + \Theta_s^{-1}(I_s - \overline{W})$ it is $|\lambda(\overline{W} + \Theta_s^{-1}(I_s - \overline{W}))| \geq 1$. Some further manipulations give the following eigenvalue localization condition: $|\lambda(I_s - (I_s - \Theta_s^{-1})(I_s - \overline{W}))| \geq 1$ which corresponds to say that the eigenvalues of $(I_s - \Theta_s^{-1})(I_s - \overline{W})$ cannot be in the interior of the unit disc centered at 1 in $\mathbb{C}$. ■

Proof of Theorem 2: Given a Schur stable matrix A, it is well known that $A - I$ is Hurwitz. According to Theorem 1, if W is ES with $1 \in \omega(W)$ a strictly dominant sharp point, then it implies $\rho(DW) < 1$ where $D = I - \Theta$.

Choose $0 < \varepsilon \leq 1 \in \mathbb{R}$ such that $\omega(\varepsilon L)$ lies strictly inside the unit circle centered at 1 with a sharp point at $0 \in \omega(\varepsilon L)$. Consequently, $W = I - \varepsilon L$ is such that $\omega(W)$ also possesses a sharp point at $1 \in \omega(W)$ and remains within the unit circle. Moreover, noticing that εL has left and right eigenvector $\mathbb{1}$ associated to eigenvalue 0, we can deduce that W has left and right eigenvector $\mathbb{1}$ associated to eigenvalue 1 and $\rho(W) = 1$ is simple, positive, and strictly dominant. Thus, W is ES, and we can apply Theorem 1. We can write

$$\rho\left((I-\Theta)(I-\varepsilon L)\right) = \rho\left(I - \Theta - \varepsilon DL\right) < 1.$$

Using the property of Schur stable matrices, $-(\Theta + \varepsilon DL)$ is Hurwitz. Following the proof of Theorem 1, we also get $r(D^{\frac{1}{2}}(I-\varepsilon L)D^{\frac{1}{2}}) < 1$, which implies that $\mathbf{x}^T(\varepsilon D^{1/2}LD^{1/2} + \Theta)\mathbf{x} > 0$. For L with a sharp point at $0 \in \omega(L)$, we have $\mathbf{x}^T L\mathbf{x} \geq 0$ and $\mathbf{x}^T D\mathbf{x} > 0$, resulting in $\mathbf{x}^T D^{1/2}LD^{1/2}\mathbf{x} \geq 0$. Since $\varepsilon > 0$, it follows that $\mathbf{x}^T(D^{1/2}LD^{1/2} + \Theta)\mathbf{x} > 0$. Consequently, $-(D^{1/2}LD^{1/2} + \Theta)$ is Hurwitz, which further implies that $-(DL + \Theta)$ is Hurwitz by applying the transformation $T = D^{1/2}$.

To show that P_L has eigenvalue 1, we can use

$$\begin{aligned} DL\mathbb{1} &= \mathbf{0} \\ (DL+\Theta)\,\mathbb{1} &= \Theta\mathbb{1} \\ \mathbb{1} &= (DL+\Theta)^{-1}\,\Theta\mathbb{1}. \end{aligned}$$

We also need to show that $\rho(P_L) = 1$ is a simple, strictly dominant, positive eigenvalue and that the left eigenvector corresponding to $\rho(P_L) = 1$ is positive (for $\theta_i > 0$ for any i) or nonnegative (when $\theta_i = 0$ for some i).

We prove the statement of the theorem by contradiction. Suppose $\rho(P_L) = \eta > 1$ with right eigenvector $\mathbf{v}$. We then obtain $(DL+\Theta)^{-1}\Theta\mathbf{v} = \eta\mathbf{v}$, which can be rewritten as: $(DL + \Theta((\eta-1)/\eta))\mathbf{v} = \mathbf{0}$, which after transformation implies that $\mathbf{x}^T(D^{1/2}LD^{1/2} + \Theta((\eta-1)/\eta))\mathbf{x} = 0$.

The above equation contradicts $\mathbf{x}^T(D^{1/2}LD^{1/2} + \Theta)\mathbf{x} > 0$, since $0 < (\eta-1)/\eta < 1$ and both $D^{1/2}LD^{1/2} \succeq 0$ and $\Theta \succeq 0$. To prove that $\rho(P_L) = 1$ is simple, notice that for $\eta = 1$, we have $\mathbf{x}^T D^{1/2}LD^{1/2}\mathbf{x} = 0$ and $corank(D^{1/2}LD^{1/2}) = 1$ as $\mu(-L) = 0$ is simple. Lastly, to prove that the left eigenvector of P_L is nonnegative, consider the left eigenvector of L, $\mathbb{1}$. We have

$$\begin{aligned} \mathbb{1}^T\,(LD+\Theta) &= \mathbb{1}^T\Theta \\ \mathbb{1}^T\Theta\,(LD+\Theta)^{-1} &= \mathbb{1}^T \\ \mathbb{1}^T\Theta\,(LD+\Theta)^{-1}\,D^{-1}\Theta &= \mathbb{1}^T D^{-1}\Theta \\ \mathbb{1}^T D^{-1}\Theta\,(DL+\Theta)^{-1}\,\Theta &= \mathbb{1}^T D^{-1}\Theta \end{aligned}$$

which proves that $\Theta D^{-1}\mathbb{1}$ is the left eigenvector of P_L corresponding to eigenvalue 1. Since $\Theta \geq 0$ and $D \geq 0$, we have $\Theta D^{-1}\mathbb{1} \geq 0$. This completes the proof. ■

Proof of Corollary 3: Denote $D = I - \Theta$, $\beta \in \omega(D)$, $\varrho \in \omega(H(L))$, $\kappa \in \omega(\Theta)$, and notice that $\omega(-\Theta - D^{\frac{1}{2}}H(L)D^{\frac{1}{2}})$ is equal to $\omega(\Re(-\Theta - D^{\frac{1}{2}}LD^{\frac{1}{2}}))$, where $-\Theta - D^{\frac{1}{2}}LD^{\frac{1}{2}}$ is similar to $-\Theta - DL$ through the transformation $T = D^{\frac{1}{2}}$. So to prove that $-\Theta - DL$ Hurwitz, it is sufficient to prove that $\mu(-\Theta - D^{\frac{1}{2}}H(L)D^{\frac{1}{2}}) < 0$. Using the property $\Lambda(-\Theta - D^{\frac{1}{2}}H(L)D^{\frac{1}{2}}) \subset -\kappa - \beta\varrho$, alongside the observations that $-\kappa < -\alpha/(1+\alpha)$, $|\beta| < 1/(1+\alpha)$, and $-\varrho \leq \alpha$, we can conclude that $\mu(-D^{\frac{1}{2}}H(L)D^{\frac{1}{2}}) \leq \alpha/(1+\alpha)$ and $\mu(-D^{\frac{1}{2}}H(L)D^{\frac{1}{2}} - \Theta) < 0$. ■

Proof of Corollary 4: Using Theorem 2, P_L is EEP for the given assumptions. To prove that P_L is a radial matrix, notice that $\omega(\varepsilon I + (1-\varepsilon)L)$ is a sharp point at ε, which results in a sharp point at ε^{-1} for $\omega((\varepsilon I + (1-\varepsilon)L)^{-1})$ and at 1 for $\omega(P_L)$ where $P_L = \varepsilon(\varepsilon I + (1-\varepsilon)L)^{-1}$. Also, notice that $0 \notin \omega(P_L)$ as $L \succeq 0$, then using the property $\Lambda(P_L^T P_L) \subset \beta^2$ with $\beta \in \omega(P_L)$ we get that $\rho(P_L^T P_L) = 1$ is a simple, positive, and strictly dominant eigenvalue. As $P_L^T P_L$ is symmetric, P_L is a radial matrix. ■

Proof of Proposition 2: Under the assumptions of the proposition, Θ_s is invertible and $(\Theta_s + (I_s - \Theta_s)\overline{L})^{-1}$ exists, and we can write $P_{\overline{L}} = (I_s - (I_s - \Theta_s^{-1})\overline{L})^{-1}$. Hence, $\rho(P_{\overline{L}}) \leq 1$ is equivalent to $|\lambda(I_s - (I_s - \Theta_s^{-1})\overline{L})| \geq 1$ for all eigenvalues of $I_s - (I_s - \Theta_s^{-1})\overline{L}$, which corresponds to say that the eigenvalues of $(I_s - \Theta_s^{-1})\overline{L}$ cannot be in the interior of the unit disc centered at 1 in $\mathbb{C}$. ■

Appendix C
Extensions

Proof of Theorem 3: In the DT case, from Theorem 1, W ES with $1 \in \omega(W)$ being a strictly dominant sharp point, means

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that $\rho(P_W) = 1$ is simple, positive, and strictly dominant. Using Lemma 9, the concatenated FJ model (10) achieves consensus. A similar argument holds for the CT case, using Theorem 2 and Lemma 9. ■

Proof of Lemma 12: Given the assumption with $D = I - \Theta$, $\rho(DW) < 1$ for the DT case and $-(DL + \Theta)$ Hurwitz for the CT case. For a normal P_W matrix in the DT case, i.e., $P_W^T P_W = P_W P_W^T$, the following steps are outlined:

$$\Theta \left(I - W^T D\right)^{-1} (I - DW)^{-1} \Theta$$
$$= (I - DW)^{-1} \Theta^2 \left(I - W^T D\right)^{-1}.$$

Replacing $D = \bar{\varepsilon} I = (1 - \varepsilon) I$ in the above expression, we get

$$(I - \bar{\varepsilon} W^T)^{-1} (I - \bar{\varepsilon} W)^{-1} = (I - \bar{\varepsilon} W)^{-1} \left(I - \bar{\varepsilon} W^T\right)^{-1}.$$

This leads to the equation

$$I - \bar{\varepsilon} W - \bar{\varepsilon} W^T + \bar{\varepsilon}^2 W W^T = I - \bar{\varepsilon} W - \bar{\varepsilon} W^T + \bar{\varepsilon}^2 W^T W,$$

which is satisfied for W normal showing that P_W is indeed normal. A similar argument exists for the CT case, completing the proof. ■

Proof of Theorem 6: Under the assumptions of the Theorem, it follows from Corollaries 2 and 4 that $P(s)$ is ES and radial. Convergence to consensus follows by observing that we can easily extend [20, Thm. 8] from normal to radial matrices. ■

The following lemma is needed in the proof of Theorem 7.

Lemma 14: Under Assumption 1 and assuming $\Theta \succ 0$, if

1) [DT case:] Assumption 6 holds,
2) [CT case:] Assumption 7 holds,

then $\overline{P}$ is a radial matrix with $\rho(\overline{P}) = 1$ where $\overline{P} \in \{\overline{P}_W, \overline{P}_L\}$ and

$$\overline{P}_W = \Theta^{1/2}(I - (I - \Theta)^{1/2} W (I - \Theta)^{1/2})^{-1} \Theta^{1/2}, \quad (20)$$

$$\overline{P}_L = \Theta^{1/2}(\Theta + (I - \Theta)^{1/2} L (I - \Theta)^{1/2})^{-1} \Theta^{1/2}. \quad (21)$$

Proof: Denote $D = I - \Theta$ and from Theorems 1 and 2, we have that P is ES with the left eigenvector $\Theta D^{-1} \mathbb{1}$. The matrix $\overline{P}$ is similar to P with the transformation matrix $T = D^{-1/2}\Theta^{1/2}$ which means that $\rho(\overline{P}) = 1$ with the associated left and right eigenvector $\Theta^{1/2} D^{-1/2} \mathbb{1}$. Furthermore, $r(\overline{P}) = 1$ as the points $\beta \in \omega(W)$ or $\tau \in \omega(L)$ map to $\varrho \in \omega(\overline{P})$ where $|\beta| \leq 1$, $\Re(\tau) \geq 0$ and $|\varrho| \leq 1$. This follows from the fact that we have $\overline{P}_L = (I + \Theta^{-1/2} D^{1/2} L \Theta^{-1/2} D^{1/2})^{-1}$, and $\overline{P}_W = (I + \Theta^{-1/2} D^{1/2} (I - W) \Theta^{-1/2} D^{1/2})^{-1}$ after some manipulation of $\overline{P}$. Here, the terms $\chi_L = \Theta^{-1/2} D^{1/2} L \Theta^{-1/2} D^{1/2}$, and $\chi_W = \Theta^{-1/2} D^{1/2} (I - W) \Theta^{-1/2} D^{1/2}$ are positive semi-definite with a sharp point at $0 \in \omega(\chi)$ where $\chi \in \{\chi_W, \chi_L\}$, which imply that $|\varrho| \leq 1$ and $0 \notin \omega(\overline{P})$. This also implies that $1 \in \omega(\overline{P})$ is a sharp point as $1 \in \partial\omega(\overline{P})$ and both $\overline{P}$ and $H(\overline{P})$ have a common eigenspace associated to the eigenvalue 1. With $0 \notin \omega(\overline{P})$, we use the property $\Lambda(\overline{P}^T \overline{P}) \subset \varrho^2$ which implies that $\rho(\overline{P}^T \overline{P}) = 1$ is simple, positive, and strictly dominant. As $\overline{P}^T \overline{P}$ is symmetric, we get $I - \overline{P}^T \overline{P} \succeq 0$ which implies that $\overline{P}$ is a radial matrix. ■

Proof of Theorem 7: To show that the time-varying concatenated SFJ model converges to consensus, we need to find a radially unbounded common Lyapunov function (CLF) for all $P(s)$, $s = 1, 2, \ldots$, i.e., to show that the time-varying model (16) is uniformly stable [20]. According to the assumptions and Lemmas 9 and 11, we have $\mathbb{1}^T W(s) = \mathbb{1}^T$ for the DT case and $\mathbb{1}^T L(s) = \mathbf{0}^T$ in the CT case for all s. Denoting $D = I - \Theta$, Theorems 1 and 2 implies $\mathbb{1}^T D^{-1} \Theta P(s) = \mathbb{1}^T D^{-1} \Theta$ and $P(s)\mathbb{1} = \mathbb{1}$ for all s. It follows that the average end-opinion of each SFJ model

$$\gamma = \gamma(s, \infty) = \gamma(s-1, \infty) = \cdots = \gamma(0, \infty) = \overline{\mathbf{w}}\mathbf{x}(s, \infty)$$
$$= \overline{\mathbf{w}}\mathbf{x}(s-1, \infty) = \cdots = \overline{\mathbf{w}}\mathbf{x}(0, \infty)$$

is a constant $\forall s$ where $\overline{\mathbf{w}} = \mathbb{1}^T D^{-1} \Theta / \|\mathbb{1}^T D^{-1} \Theta\|_1$. Hence, as in the standard consensus problem [32], we can consider the "disagreement" projection vector on $\mathrm{span}(\mathbb{1})^\perp$, $\delta(s, \infty) = \mathbf{x}(s, \infty) - \gamma \mathbb{1}$, which obeys the following equation:

$$\delta(s+1, \infty) = P(s)\mathbf{x}(s, \infty) - \gamma \mathbb{1} = P(s)\delta(s, \infty). \quad (22)$$

From now on we omit the second index in $\mathbf{x}(s, \infty)$, $\delta(s, \infty)$, and $\gamma(s, \infty)$ for simplicity of the notation. The function $V(s) = \delta^T(s) \Theta D^{-1} \delta(s)$ is a radially unbounded CLF for the dynamics (22). Let us consider $P(s) = P_L(s)$ for the CT case. We can write $\Delta V(s) = V(s+1) - V(s)$ as

$$\Delta V(s) = \delta^T(s) \left(P_L^T(s) \Theta D^{-1} P_L(s) - \Theta D^{-1}\right) \delta(s)$$
$$= \delta^T(s) \Big(\Theta \left(\Theta + L^T(s) D\right)^{-1} \Theta D^{-1}$$
$$(\Theta + DL(s))^{-1} \Theta - \Theta D^{-1} \Big) \delta(s)$$
$$= \delta^T(s) (D^{-1/2} \Theta \left(\Theta + D^{1/2} L^T(s) D^{1/2}\right)^{-1} \Theta$$
$$\left(\Theta + D^{1/2} L(s) D^{1/2}\right)^{-1} \Theta D^{-1/2} - \Theta D^{-1}) \delta(s).$$

Denoting $\tilde{\delta}(s) = D^{-1/2} \Theta^{1/2} \delta(s)$ where $D^{-1/2} \Theta^{1/2} \succ 0$, we obtain

$$\Delta V(s) = \tilde{\delta}^T(s) \Big(\Theta^{1/2} \left(\Theta + D^{1/2} L^T(s) D^{1/2}\right)^{-1} \Theta$$
$$\left(\Theta + D^{1/2} L(s) D^{1/2}\right)^{-1} \Theta^{1/2} - I \Big) \tilde{\delta}(s)$$
$$= \tilde{\delta}^T(s) (\overline{P}_L^T(s) \overline{P}_L(s) - I) \tilde{\delta}(s).$$

Similarly, for the DT case, choosing $P(s) = P_W(s)$, $\Delta V(s) = V(s+1) - V(s)$ can be written as

$$\Delta V(s) = \tilde{\delta}^T(s) \Big(\Theta^{1/2} \left(I - D^{1/2} W^T(s) D^{1/2}\right)^{-1} \Theta$$
$$\left(I - D^{1/2} W(s) D^{1/2}\right)^{-1} \Theta^{1/2} - I \Big) \tilde{\delta}(s)$$
$$= \tilde{\delta}^T(s) \left(\overline{P}_W^T(s) \overline{P}_W(s) - I\right) \tilde{\delta}(s).$$

Using Lemma 14, $\overline{P}_L(s)$ and $\overline{P}_W(s)$ are radial matrices. Denote $\overline{P}(s) = \{\overline{P}_L(s), \overline{P}_W(s)\}$ for the respective CT and DT cases. This gives us that $\Lambda(\overline{P}^T(s) \overline{P}(s)) = \{\nu_1(s), \ldots, \nu_{n-1}(s), 1\}$, where $\nu_i(s) \in \mathbb{R}$ with $|\nu_1(s)| \leq \cdots \leq |\nu_{n-1}(s)| < 1$. Notice that the right and left eigenvector of $\overline{P}(s)$ associated to eigenvalue 1 is $D^{-1/2} \Theta^{1/2} \mathbb{1}$ $\forall s$. Then, since $\tilde{\delta}(s) \perp$

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$\mathrm{span}(D^{-1/2}\Theta^{1/2}\mathbb{1})$ (i.e., it is $\tilde{\delta}(s) \in \mathrm{span}(D^{-1/2}\Theta^{1/2}\mathbb{1})^{\perp}$), we can apply the Courant–Fischer theorem and it holds that $\Delta V(s) \leq (\tilde{\lambda} - 1)V(s)$ where $\tilde{\lambda} = \max |\nu_{n-1}(s)| < 1$. Therefore, $V(s+1) \leq \tilde{\lambda} V(s)$, and $V(s+1) \leq \tilde{\lambda}^s V(0)$ which yields $\lim_{s\to\infty} \delta(s) = 0$. This completes the proof. ■

Proof of Theorem 8: From Theorems 1 and 2, $P(s)$ is ES $\forall s$. This means that the projection of $P(s)$ onto $\mathrm{span}(\mathbb{1})^{\perp}$, i.e., $\tilde{P}(s) = \Gamma P(s) \Gamma^T$, is Schur stable. Denote $\tilde{Q} = \Gamma Q \Gamma^T$ the projection of Q on $\mathrm{span}(\mathbb{1})^{\perp}$, then $Q \succ 0$ implies $\tilde{Q} \succ 0$ and $\tilde{P}^T(s)\tilde{Q}\tilde{P}(s) - \tilde{Q} = \Gamma(P^T(s)QP(s) - Q)\Gamma^T \prec 0$ implies (19). ■

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Muhammad Ahsan Razaq received the bachelor's degree in electrical engineering from the National University of Sciences and Technology, Islamabad, Pakistan, in 2016, and the master's degree in systems engineering from the Pakistan Institute of Engineering and Applied Sciences, Islamabad, in 2018. He is currently working toward the doctoral degree in the Division of Automatic Control with the Department of Electrical Engineering, Linköping University, Linköping, Sweden.

His research focuses on multiagent systems with applications to opinion dynamics.

Claudio Altafini (Senior Member, IEEE) received the master's degree ("Laurea") in electrical engineering from the University of Padova, Padua, Italy, in 1996, and the Ph.D. degree in optimization and systems theory from the Royal Institute of Technology, Stockholm, Sweden in 2001.

From 2001 to 2013, he was with the International School for Advanced Studies, Trieste, Italy. Since 2014, he is a Professor in the Division of Automatic Control with the Department of Electrical Engineering, Linköping University, Linköping, Sweden. His research interests are in the areas of nonlinear control and multiagent systems, with applications to social networks, systems biology, quantum mechanics, and complex networks in general.

Dr. Altafini was an Associate Editor for IEEE Transactions on Automatic Control, IEEE Transactions on Control of Network Systems, and *Automatica*.

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